

In the end the world will be a desert of chairs and sofas ... rolling through infinity with no-one to sit on them.

E. M. Forster, *Howards End*

10.1 Brief description of the Universe

Our Sun is one star in a collection of about 10^{11} stars forming our Galaxy. The Galaxy is shaped roughly like a pancake – approximately circular in ‘plan’ and with thickness much less than its radius – and the Sun is situated towards the outside of this distribution, not far from the central plane. The Galaxy is about 100 000 light years (ly) across. Almost all the stars visible to the naked eye at night belong to our Galaxy, and looking at the Milky Way is looking into its central plane, where the density of stars is greatest. The Andromeda Nebula, also visible to the naked eye, is a separate galaxy about 2 million light years away, and in fact is a member of the Local Group of galaxies. The construction of large telescopes in the first decades of the twentieth century led to the discovery of many galaxies and groups of galaxies and it is now known that there are about 10^{11} galactic clusters in the visible Universe. Considering these clusters as the ‘elementary’ constituents of the Universe, on scales larger than that of the clusters their distribution in space appears to be *homogeneous and isotropic*. This is the first – and very remarkable – feature of the Universe.

An important cosmological figure is the average density of matter in the Universe. The contribution of the matter contained in galaxies is

$$\rho_b \approx 10^{-31} \text{ g cm}^{-3} = 10^{-28} \text{ kg m}^{-3}. \quad (10.1)$$

The subscript b refers to the fact that the ‘matter’ referred to here is *baryonic* matter – made of protons and neutrons. This is the sort of matter of which stars and planets and we ourselves are made. We shall see in due course that there is matter of a different sort – dark matter – in the Universe. The matter referred to in (10.1) is not dark; it constitutes stars, and shines.

The discovery that the Universe is isotropic means that it has *no centre* – there is no privileged point in space. The space (not space-time) of the Universe is a homogeneous space. This is the content of the so-called Cosmological Principle (CP). An extension of this principle is the Perfect Cosmological Principle (PCP) according to which, if it were true, the

Universe would be homogeneous in space *and in time*. It would look the same at all points in space and at all times, which of course implies that it must be of infinite age, in both the past and the future. This principle was proposed by Hermann Bondi and Thomas Gold in 1948 and came to be known as the Steady State Theory.¹ It might be thought that Special Relativity requires that the statement of homogeneity in space be extended to homogeneity in space and time, but this is *not true*, since Special Relativity is a theory about the laws of nature. It demands that the laws of nature be covariant – have the same form – under Lorentz transformations. In doing physics we apply the laws of nature to a particular physical situation – for example studying the motion of a charged particle in an electric field. If the laws of nature are ultimately differential equations – and they are – the situation to which we apply these laws (the motion of the particle) is represented by the *boundary conditions* of these equations (the initial position and speed of the particle). It is indeed a most remarkable circumstance that our knowledge can be divided into these two categories.² In the present case the physical system is the Universe as a whole, which therefore plays the role of a ‘boundary condition’, so the Cosmological Principle is a property of the boundary conditions, not a property of any law of nature, and Special Relativity does not require that it should be extended to a Perfect Cosmological Principle. In fact the Steady State Theory is now discredited; in Big Bang Cosmology the Universe certainly looked very different in the past from how it looks now!

One of the major advances in cosmology was the discovery by Edwin Hubble in the 1920s, working on the big telescopes at the Mount Wilson and Palomar Observatories, that the Universe is *expanding*. This was his interpretation of the fact that the light from far galaxies was *red-shifted* relative to light from nearby galaxies. Hubble interpreted the red-shift as a Doppler shift and concluded that the galaxies are moving away – from us, and from each other. The observed pattern of expansion was that the galaxies are moving away radially, with a speed v proportional to their distance r from us,

$$v = H_0 r. \quad (10.2)$$

The constant H_0 is the present value of *Hubble’s constant*, sometimes called the *Hubble parameter*, since it is likely that H_0 varies in time. There is a degree of uncertainty in its present value, which is

$$H_0 = (55\text{--}85) \text{ km s}^{-1} \text{ Mpc}^{-1}; \quad (10.3)$$

a galaxy 1 Mpc (= 3.3 ly) away is receding at a speed of around 70 km s^{-1} . It is convenient to write (10.3) as

$$H_0 = h \cdot 100 \text{ km s}^{-1} \text{ Mpc}^{-1};$$

with h at present in the range 0.55 to 0.85.

There are a number of observations to make about this important discovery. Firstly, it is not a particularly rapid expansion. Since $1 \text{ Mpc} = 3.09 \times 10^{19} \text{ km}$ and $1 \text{ year} = 3.15 \times 10^7 \text{ s}$, then, on putting $h = 1$ for definiteness, a galaxy 10^{20} km (about 3 million ly) distant from us

¹ For an account of the Steady State Theory see for example Bondi (1960).

² These observations are taken from Wigner’s essay ‘Symmetry and conservation laws’, in Wigner (1967).



Fig. 10.1

Hubble expansion in one dimension: everyone sees the same pattern.

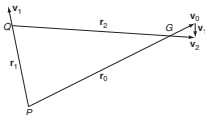


Fig. 10.2

Hubble expansion in three dimensions.

will have receded, after 10^8 years, by about 10^{18} km – an expansion of only 1% in 100 million years. In units of inverse seconds H_0 is

$$H_0 = h (3.24 \times 10^{-18}) \text{ s}^{-1}$$

whose inverse is

$$H_0^{-1} = \frac{3.09}{h} \times 10^{17} \text{ s} \approx \frac{1}{h} \times 10^{10} \text{ years.} \quad (10.4)$$

If H_0 were a constant in time this figure would represent the age of the Universe – the time that has elapsed since it had ‘zero’ size. In any case, even if Hubble’s parameter is time dependent, its inverse should certainly bear some relation to the age of the Universe.

A second observation is that Hubble’s law (10.2) does *not* imply that we are at the centre of expansion (the centre of the Universe), as initially one might be tempted to think. Consider the expansion in one dimension. Suppose we have points M , N , P , R and S a distance d apart from each other (see Fig. 10.1). An observer at P sees himself at rest, sees an observer at R moving to the right with speed $v = Hd$, an observer at S moving to the right with speed $2v$, an observer at N moving to the left with speed v and so on. An observer at R , on the other hand, will see herself at rest, will see S moving to the right with speed v , P moving to the left with speed v , N to the left with speed $2v$, and so on; she sees exactly the same pattern of expansion as P does. Every observer sees the same, so there is no centre of expansion. It is easy to generalise this to three dimensions. Suppose an observer at P sees an arbitrary galaxy G , a distance $\mathbf{r}_0$ away, receding (radially) at speed $\mathbf{v}_0 = H\mathbf{r}_0$, as in Fig. 10.2. What does an observer at Q see? Q moves at velocity $\mathbf{v}_1 = H\mathbf{r}_1$ with respect to P , so by vector addition of velocities will see G , a distance $\mathbf{r}_2$ away, move with velocity

$$\mathbf{v}_2 = \mathbf{v}_0 - \mathbf{v}_1 = H(\mathbf{r}_0 - \mathbf{r}_1) = H\mathbf{r}_2;$$

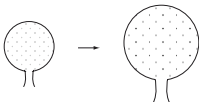


Fig. 10.3

Dots on a balloon move apart as the balloon inflates.

Q sees a Hubble expansion, just as P does – everybody sees the same thing. It is clear that Hubble's law (10.2) is the only expansion law compatible with a homogeneous universe: if Hubble had observed that v were proportional to r^2 then we would certainly have been at the expansion centre, and the hypothesis of a homogeneous Universe would have been ruined.

Our third observation involves gravity. Since all the galaxies attract one another (in Newtonian language) this will have the effect of *slowing down* the expansion – we would expect to observe a *deceleration* in the motion of the far galaxies. It has been clear in recent years, however, that the expansion is accelerating. We shall see, in the course of this chapter, that General Relativity, with the Einstein field equations (5.24) also predicts a decelerating universe. To account for an accelerating one involves introducing a 'cosmological constant' Λ into the field equations. We return to this below.

The fourth and final observation is slightly more subtle. It is the statement that the expansion observed by Hubble is not to be interpreted as the motion of galaxies through a fixed, static, space. It is, rather, the observation that the *space itself* is expanding, carrying the galaxies with it. We may visualise the analogy of a balloon being inflated, as in Fig. 10.3. If there are dots painted on the rubber surface of the balloon, then as it is inflated the dots move further apart because they are embedded in the rubber, which is itself stretching. So, on the cosmological scale, in General Relativity, it is the expansion of space itself which is being manifested. After all, in Einstein's vision, space has a reality that it never possessed in physics before. It acquires *curvature* when masses are placed in it; and now we learn that on the scale of the Universe as a whole it is expanding.

10.2 Robertson-Walker metric

Our first task is to translate the Cosmological Principle, that the Universe is homogeneous and isotropic, into an explicitly geometrical condition to be satisfied by the space-time metric – more particularly by its space part. We use a method based on Killing vectors, introduced in Section 6.6. It was shown there that if a space (or space-time) possesses a particular symmetry then the Lie derivative of the Killing vector generating the symmetry vanishes; that is, from (6.133)

$$g_{\mu\nu,\lambda}\xi^\lambda + g_{\mu\lambda}\xi^\lambda_{,\nu} + g_{\lambda\nu}\xi^\lambda_{,\mu} = 0, \quad (10.5)$$

where ξ^λ is the relevant Killing vector. Let us first implement the requirement of isotropy – that the metric is invariant under spatial rotations. In fact, we have already written down the most general form of *static* metric with spherical symmetry, which is (5.31)

$$ds^2 = -e^{2\nu} c^2 dt^2 + e^{2\lambda} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (10.6)$$

where $\nu = \nu(r)$, $\lambda = \lambda(r)$; this was a preliminary step to finding the Schwarzschild solution. In the case of cosmology there is no requirement that the metric be static; we shall proceed to implement the requirement of isotropy only, followed by the requirement of homogeneity. We recall from the (6.138) that the Killing vectors for rotations in space are

$$X_4 = -z \frac{\partial}{\partial y} + y \frac{\partial}{\partial z}, \quad X_5 = z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}, \quad X_6 = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}.$$

Changing notation and writing these as 4-vectors rather than 3-vectors, the Cartesian components of the Killing vectors are

$$\xi^\mu_1 = (0, 0, -z, y), \quad \xi^\mu_2 = (0, z, 0, -x), \quad \xi^\mu_3 = (0, -y, x, 0). \quad (10.7)$$

Then for example, transforming to spherical polars,

$$\begin{aligned} \xi_1 &= -z \frac{\partial}{\partial y} + y \frac{\partial}{\partial z} \\ &= -r \cos \theta \left(\sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \sin \phi \frac{\partial}{\partial \theta} + \frac{1}{r} \frac{\cos \phi}{\sin \theta} \frac{\partial}{\partial \phi} \right) \\ &\quad + r \sin \theta \cos \phi \left(\cos \theta \frac{\partial}{\partial r} - \sin \theta \frac{\partial}{\partial \theta} \right) \\ &= -\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi}; \\ \xi^\mu_1 &= (0, 0, -\sin \phi, -\cot \theta \cos \phi), \end{aligned} \quad (10.8)$$

where in the last line the components are in spherical polar coordinates. Similarly we find

$$\xi^\mu_2 = (0, 0, \cos \phi, -\cot \theta \sin \phi), \quad \xi^\mu_3 = (0, 0, 0, 1). \quad (10.9)$$

To find the metric tensor we use Killing's equation (10.5). Applied to ξ^μ_3 it gives

$$g_{\mu\nu,3} = 0; \quad \frac{\partial}{\partial \phi} g_{\mu\nu} = 0. \quad (10.10)$$

Putting $(\mu\nu) = (00)$, (10.5) applied to ξ_1 gives

$$\begin{aligned} g_{00,2}(-\sin \phi) + g_{00,3}(-\cot \theta \cos \phi) &= 0, \\ g_{00,2} \sin^2 \phi &= -g_{00,3} \cot \theta \cos \phi \sin \phi, \end{aligned}$$

and applied to ξ_2 gives

$$g_{00,2} = 0; \quad \frac{\partial}{\partial \theta} g_{00} = 0. \quad (10.11)$$

Similarly, for $(\mu\nu) = (01)$, (11), since ξ_1 and ξ_2 are independent of t and r we find

$$g_{10,2} = 0, \quad g_{11,2} = 0. \quad (10.12)$$

In the same vein, with $(\mu\nu) = (22), (23), (33), (12), (13), (02), (03)$, the Killing vector equation for ξ_1^μ gives the seven equations

$$g_{22,2} \sin \phi = -2g_{23} \frac{\cos \phi}{\sin^2 \theta}, \quad (10.13)$$

$$(g_{23,2} - g_{23} \cot \theta) \sin \phi = \left(-g_{22} + \frac{g_{33}}{\sin^2 \theta} \right) \cos \phi, \quad (10.14)$$

$$(-g_{33,2} + g_{33} \cot \theta) \sin \phi = 2g_{32} \cos \phi, \quad (10.15)$$

$$g_{12,2} \sin \phi = -2g_{13} \frac{\cos \phi}{\sin^2 \theta}, \quad (10.16)$$

$$(-g_{13,2} + g_{13} \cot \theta) \sin \phi = g_{12} \cos \phi, \quad (10.17)$$

$$g_{02,2} \sin \phi = g_{03} \frac{\cos \phi}{\sin^2 \theta}, \quad (10.18)$$

$$(g_{03,2} - g_{03} \cot \theta) \sin \phi = g_{02} \cos \phi. \quad (10.19)$$

To these seven equations are to be added the seven similar equations for ξ_2^μ , which actually may be found by making the substitutions $\sin \phi \rightarrow -\cos \phi$, $\cos \phi \rightarrow \sin \phi$; hence

$$g_{22,2} \cos \phi = 2g_{23} \frac{\sin \phi}{\sin^2 \theta}, \quad (10.20)$$

$$-(g_{23,2} - g_{23} \cot \theta) \cos \phi = \left(-g_{32} + \frac{g_{33}}{\sin^2 \theta} \right) \sin \phi, \quad (10.21)$$

$$(g_{33,2} - g_{33} \cot \theta) \cos \phi = 2g_{32} \sin \phi, \quad (10.22)$$

$$-g_{12,2} \cos \phi = -2g_{13} \frac{\sin \phi}{\sin^2 \theta}, \quad (10.23)$$

$$(g_{13,2} - g_{13} \cot \theta) \cos \phi = g_{12} \sin \phi, \quad (10.24)$$

$$-g_{02,2} \cos \phi = g_{03} \frac{\sin \phi}{\sin^2 \theta}, \quad (10.25)$$

$$-(g_{03,2} - g_{03} \cot \theta) \cos \phi = g_{02} \sin \phi. \quad (10.26)$$

Gathering our results together, (10.10), (10.11) and (10.12) give

$$g_{00} = g_{00}(t, r), \quad g_{01} = g_{01}(t, r), \quad g_{11} = g_{11}(t, r). \quad (10.27)$$

Equations (10.13) and (10.20) give $g_{23} = 0$, $g_{22,2} = 0$, which together with (10.10) implies

$$g_{22} = g_{22}(t, r), \quad g_{23} = 0. \quad (10.28)$$

This last condition, together with (10.14) gives

$$g_{33} = g_{22}(t, r) \sin^2 \theta. \quad (10.29)$$

Finally, (10.18) and (10.25) together give $g_{03} = 0$, $g_{02,2} = 0$, hence from (10.26) $g_{02} = 0$; and (10.17) and (10.24) give $g_{12} = 0$, hence (10.23) gives $g_{13} = 0$ and we conclude that $g_{\mu\nu}$ has the form

$$g_{\mu\nu} = \begin{pmatrix} g_{00}(r, t) & g_{01}(r, t) & 0 & 0 \\ g_{10}(r, t) & g_{11}(r, t) & 0 & 0 \\ 0 & 0 & g_{22}(r, t) & 0 \\ 0 & 0 & 0 & g_{22}(r, t) \sin^2 \theta \end{pmatrix}, \quad (10.30)$$

or

$$ds^2 = -g_{00}(r, t) c^2 dt^2 + 2g_{01}(r, t) c dr dt + g_{11}(r, t) dr^2 + g_{22}(r, t) [d\theta^2 + \sin^2 \theta d\phi^2]. \quad (10.31)$$

This is the most general metric for an *isotropic* space. On the cosmological scale, however, space is also *homogeneous*. A space is homogeneous if it is isotropic about one point and invariant under *translations*; this clearly amounts to being isotropic about all points. The Killing vectors for translations are, from (6.138), in Cartesian coordinates

$$\eta_1 = \frac{\partial}{\partial x}, \quad \eta_1^\mu = (0, 1, 0, 0),$$

or, in spherical polars

$$\begin{aligned} \eta_1 &= \sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}, \\ \eta_1^\mu &= \left(0, \sin \theta \cos \phi, \frac{1}{r} \cos \theta \cos \phi, -\frac{1}{r \sin \theta} \right). \end{aligned} \quad (10.32)$$

Similarly

$$\begin{aligned} \eta_2^\mu &= \left(0, \sin \theta \sin \phi, \frac{1}{r} \cos \theta \sin \phi, \frac{1}{r \sin \theta} \right), \\ \eta_3^\mu &= \left(0, \cos \theta, -\frac{1}{r} \sin \theta, 0 \right). \end{aligned} \quad (10.33)$$

Killing's equation (10.5) applied to (10.33) gives, for $(\mu\nu) = (00)$

$$g_{00,1} = 0 \Rightarrow g_{00} = g_{00}(t), \quad (10.34)$$

and with $(\mu\nu) = (22)$

$$g_{22}(r, t) = A(t) r^2. \quad (10.35)$$

Finally $(\mu\nu) = (12)$ and (02) give

$$g_{11} = \frac{1}{f^2} g_{22} = A(t), \quad g_{01} = 0$$

and these conditions inserted into (10.31) give a line element

$$ds^2 = -g_{00}(t) c^2 dt^2 + A(t)[dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2]. \quad (10.36)$$

Under a transformation $t \rightarrow t'$ with $dt'^2 = g_{00}(t) dt^2$, this may be written, dropping the prime,

$$ds^2 = -c^2 dt^2 + S^2(t)[dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2]. \quad (10.37)$$

In this cosmological metric t is a 'universal' time, defined over the whole space-time manifold; it may be identified with proper time at each galaxy (since all galaxies are equivalent). The factor $S^2(t)$ is clearly an 'expansion' factor, affecting the spatial part of the metric, and therefore indicating an expanding universe. We have then proved rather formally that an expanding universe is compatible with a homogeneous and isotropic one – the fact that there is an expansion does *not* mean that there is a 'centre' of expansion. The common analogy, already described, is with the dots painted on the surface of a balloon which is then inflated – see Fig. 10.3 above.

The spatial section in the line element above, however, is *flat*; including the expansion factor $S^2(t)$ it is *conformally* flat. We must generalise this and consider the cases where it is curved, which, in general, it may be. A curved spatial hypersurface, however, may be embedded into a flat 4-dimensional Euclidean space E^4 , just as the spherical surface S^2 may be considered embedded into the flat space E^3 . The simplest model is a simple generalisation of this case; that the spatial section is essentially a 3-sphere S^3 of radius a with the equation

$$(x^1)^2 + (x^2)^2 + (x^3)^2 + (x^4)^2 = a^2, \quad (10.38)$$

where $x^1, \dots, x^4$ are the Cartesian coordinates in E^4 . This equation is analogous to that of S^2 in E^3 : $x^2 + y^2 + z^2 = a^2$ is a spherical surface of radius a . The fourth coordinate x^4 above has, of course, nothing to do with time; E^4 is a ('fictitious') flat space in which is embedded 3-physical-dimensional space. It is convenient to introduce spherical coordinates for S^3 ,

$$\begin{aligned} x^1 &= a \sin \chi \sin \theta \cos \phi, & x^2 &= a \sin \chi \sin \theta \sin \phi, \\ x^3 &= a \sin \chi \cos \theta, & x^4 &= a \cos \chi. \end{aligned} \quad (10.39)$$

The coordinates χ, θ, ϕ label points on S^3 , so we are now using a coordinate system

$$(x^0, x^1, x^2, x^3) = (ct, \chi, \theta, \phi). \quad (10.40)$$

In dealing with the flat space E^3 above we found that the Killing vectors for rotations, (10.8), (10.9), gave us, via Killing's equation (10.5), the isotropic line element (10.30). In the present case we have a slightly different coordinate system so we should check that our new Killing vectors also give an isotropic metric. Analogous to (10.7) the three Killing vectors for rotations in the (23), (31) and (12) planes are

$$\xi_1 = x^2 \frac{\partial}{\partial x^3} - x^3 \frac{\partial}{\partial x^2}, \quad \xi_2 = x^3 \frac{\partial}{\partial x^1} - x^1 \frac{\partial}{\partial x^3}, \quad \xi_3 = x^1 \frac{\partial}{\partial x^2} - x^2 \frac{\partial}{\partial x^1}. \quad (10.41)$$

In the coordinate system (10.40) we find

$$\begin{aligned}\xi_1^\mu &= \sin \chi \sin \theta \sin \phi \left(0, \cos \chi \cos \theta, -\frac{\sin \theta}{\sin \chi}, 0 \right) \\ &\quad - \sin \chi \cos \theta \left(0, \cos \chi \sin \theta \sin \phi, \frac{\cos \theta \sin \phi}{\sin \chi}, \frac{\cos \phi}{\sin \chi \sin \theta} \right) \\ &= (0, 0, -\sin \phi, -\cot \theta \cos \phi),\end{aligned}$$

exactly as in (10.8). The other Killing vectors are also given by (10.9). The consequences of isometry under these rotations therefore follow exactly from the equations above and we end up with the metric (10.30), or

$$\begin{aligned}ds^2 &= -g_{00}(\chi, t) c^2 dt^2 + 2g_{01}(\chi, t) c d\chi dt \\ &\quad + g_{11}(\chi, t) d\chi^2 + g_{22}(\chi, t) [d\theta^2 + \sin^2 \theta d\phi^2],\end{aligned}\quad (10.42)$$

the coordinate χ playing the role of r in the flat case.

We must now extend the symmetry to include isometry under rotations in the (14), (24) and (34) planes. These take the place of translations in the flat case, since there is no translational symmetry in a curved space – see the examples at the end of Section 6.6. The Killing vectors for rotations in the (14), (24) and (34) planes are

$$\eta_1 = x^4 \frac{\partial}{\partial x^1} - x^1 \frac{\partial}{\partial x^4}, \quad \eta_2 = x^4 \frac{\partial}{\partial x^2} - x^2 \frac{\partial}{\partial x^4}, \quad \eta_3 = x^4 \frac{\partial}{\partial x^3} - x^3 \frac{\partial}{\partial x^4}. \quad (10.43)$$

The components of η_3 in the basis $\left(\frac{\partial}{\partial t}, \frac{\partial}{\partial \chi}, \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \phi} \right)$ are

$$\begin{aligned}\eta_3^\mu &= -\sin \chi \cos \theta (0, -\sin \chi, 0, 0) + \cos \chi \left(0, \cos \chi \cos \theta, -\frac{\sin \theta}{\sin \chi}, 0 \right) \\ &= (0, \cos \theta, -\cot \chi \sin \theta, 0),\end{aligned}\quad (10.44)$$

or

$$\eta_3 = \cos \theta \frac{\partial}{\partial \chi} - \cot \chi \sin \theta \frac{\partial}{\partial \theta}. \quad (10.45)$$

Killing's equation (10.5) for η_3^μ gives, for $(\mu\nu) = (00)$,

$$g_{00,1} \cos \theta = 0,$$

hence

$$g_{00} = g_{00}(t). \quad (10.46)$$

For $(\mu\nu) = (22)$ we find

$$g_{22,1} = 2g_{22} \cot \chi,$$

which on integration gives

$$g_{22} = B(t) \sin^2 \chi. \quad (10.47)$$

Putting $(\mu\nu) = (12)$ gives

$$g_{22} = g_{11} \sin^2 \chi, \quad (10.48)$$

which with (10.47) yields

$$g_{11} = g_{11}(t). \quad (10.49)$$

Finally, $(\mu\nu) = (02)$ gives

$$g_{01} = 0,$$

hence (see (10.30)) $g_{\mu\nu}$ is *diagonal*,

$$g_{\mu\nu} = \text{diag}\{g_{00}(t), g_{11}(t), \sin^2\chi g_{11}(t), \sin^2\chi \sin^2\theta g_{11}(t)\},$$

or, redefining t , as before,

$$ds^2 = -c^2 dt^2 + S^2(t)[d\chi^2 + \sin^2\chi d\theta^2 + \sin^2\chi \sin^2\theta d\phi^2], \quad (10.50)$$

where $S(t)$ is some function of t . Observing that $\sin^2\chi = r^2/a^2$, then

$$d\chi^2 = \frac{dr^2}{r^2 - a^2}$$

and, absorbing a factor of $1/a^2$ into $S^2(t)$ we have

$$ds^2 = -c^2 dt^2 + S^2(t) \left(\frac{dr^2}{1 - \frac{r^2}{a^2}} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right). \quad (10.51)$$

Instead of the embedding (10.38) of cosmological 3-space into E^4 , however, we could have the embedding

$$(x^4)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2 = a^2, \quad (10.52)$$

which describes a *hyperbolic* space. It may be coordinatised by

$$\begin{aligned} x^1 &= a \sinh \chi \sin \theta \cos \phi, & x^2 &= a \sinh \chi \sin \theta \sin \phi, \\ x^3 &= a \sinh \chi \cos \theta, & x^4 &= a \cosh \chi, \end{aligned} \quad (10.53)$$

which is just (10.39) but with $\sin \chi \rightarrow \sinh \chi$, $\cos \chi \rightarrow \cosh \chi$. The Killing vectors may be written down and the analysis goes through very similarly to that above, finishing up with the metric

$$ds^2 = -c^2 dt^2 + R^2(t) \left(\frac{dr^2}{1 + \frac{r^2}{a^2}} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right) \quad (10.54)$$

to replace (10.50). This is the cosmological space-time metric for a homogeneous, isotropic universe with a hyperbolic 3-space section. The possible metrics, (10.37), (10.51) and (10.54), may be expressed in the single form

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{1}{1 - Kr^2} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right], \quad (10.55)$$



Fig. 10.4

The intersection of a plane with a surface: at a point of intersection the vector $\mathbf{v}_1$ is tangent to a circle of radius r_1 .

with

$$K = 0, \quad \frac{1}{a^2} \quad \text{or} \quad -\frac{1}{a^2}, \quad (10.56)$$

respectively. The quantity K is called the *Gaussian curvature*. This quantity is defined for (2-dimensional) surfaces in a 3-dimensional space, in the following way. Consider a 2-dimensional surface, and visualise the intersection of a plane with this surface in some region, as in Fig. 10.4. At a point P on the line of intersection a tangent vector $\mathbf{v}_1$ may be drawn, and in the direction of $\mathbf{v}_1$ the intersection of the surface and the plane defines, to lowest order, a *circle*, with radius say r_1 . We then say that the curvature of the surface at P in the direction of the tangent vector $\mathbf{v}_1$ is $k_1 = \frac{1}{r_1}$ (large radius, small curvature and vice versa). In a direction *perpendicular* to $\mathbf{v}_1$ the same is true: in the direction of a tangent vector $\mathbf{v}_2$ the intersection defines a circle with radius r_2 , say, and curvature $k_2 = \frac{1}{r_2}$. The Gaussian curvature K at P is

$$K = k_1 k_2 = \frac{1}{r_1 r_2}. \quad (10.57)$$

Gauss showed that K does *not* depend on the choice of the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, or equivalently on the planes of intersection. In this sense K is an *invariant* of the surface at the point considered. It may, however, have a positive or negative sign. Consider, for example, a sphere S^2 of radius a . Because S^2 is a homogeneous space (like S^3 is) the curvature is the same everywhere, so that at any point on the sphere both k_1 and k_2 are equal to $1/a$ and

$$S^2: \quad K = \frac{1}{a^2} > 0; \quad (10.58)$$

the Gaussian curvature is positive. This is because at any point on the sphere the centres of the two circles (defining the curvatures in two perpendicular directions) are on the *same side* of the surface ('inside' it). In contrast, consider a point on the surface of a hyperboloid, as in Fig. 1.8. It is clear that if the plane of intersection with this surface is (zy) the centre of the circle of intersection is *outside* the surface, but the circle resulting from the intersection with the (xz) plane is *inside* it. Since the centres are on opposite sides of the surface the Gaussian curvature is negative,

$$\text{hyperboloid:} \quad K < 0. \quad (10.59)$$

A flat surface (plane), of course, has $K = 0$. In these examples we have considered surfaces (curved or flat) embedded in a *flat* Euclidean space E^3 . In the context of cosmology we are considering the Gaussian curvature K of a 3-dimensional space, not a 2-dimensional one, but actually K , or a generalisation of it, may be defined for a space of any dimension. In fact the definition is given in Equation (10.63) below.

It is common to simplify the metric (10.55) by expressing it solely in terms of the *sign* of the Gaussian curvature, rather than the actual magnitude of it. Define

$$k = \frac{K}{|K|} \quad (10.60)$$

so that

$$\begin{aligned} k = 1 & \quad \text{closed space } (S^3), \\ k = 0 & \quad \text{flat space } (E^3), \\ k = -1 & \quad \text{open space (hyperbolic)}. \end{aligned} \quad (10.61)$$

Then defining $r' = \sqrt{K}r$, $\frac{R^2(t)}{K} = S^2(t)$ and dropping the prime, (10.55) becomes

$$ds^2 = -c^2 dt^2 + S^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right). \quad (10.62)$$

In this form the metric is known as the *Robertson–Walker metric*, after its original authors.³ It has been obtained without the use of the field equations, appealing only to the symmetry requirements of isotropy and homogeneity. The factor $S^2(t)$ describes an *expanding* space (or, in principle, a contracting one, though that is not what our Universe is doing at present) and it is noteworthy that this feature arises naturally from the above symmetry requirements. The coordinates above are *comoving*; that is, they move with the matter and are invariant along its world-lines.

It is of great interest to know what the function $S(t)$ is, in the actual Universe, and also to know what k is. As we shall see, these questions may be answered when we consider the *field equations*, which we have not yet done. It is worth noting, however, at this point, that the steps we have taken so far may be expressed in a slightly different way. It was noted in Chapter 6 that a space of dimension n is called a *maximally symmetric space* if it supports $n(n+1)/2$ Killing vectors. We have seen that imposing the conditions of isotropy and homogeneity implied that the 3-space of the universe allows six Killing vectors, so it follows that this space is a maximally symmetric space, which is also called a space of *constant curvature*. Such a space is characterised by the property⁴

$$R_{\kappa\lambda\mu\nu} = K(g_{\kappa\mu}g_{\lambda\nu} - g_{\kappa\nu}g_{\lambda\mu}), \quad (10.63)$$

where K is the Gaussian curvature, a generalisation of what was discussed above for surfaces in a 3-dimensional space. This quantity appears because at any point P of a manifold we may define the *sectional curvature*⁵

³ Robertson (1935, 1936), Walker (1936).

⁴ See for example Weinberg (1972), chapter 13.

⁵ See for example Eisenhart (1926).

$$K_P = \frac{R_{\kappa\lambda\mu\nu} v_1^\kappa v_2^\lambda v_1^\mu v_2^\nu}{(g_{\kappa\mu} g_{\lambda\nu} - g_{\kappa\nu} g_{\lambda\mu}) v_1^\kappa v_2^\lambda v_1^\mu v_2^\nu}, \quad (10.64)$$

where v_1^μ and v_2^μ are two orthogonal vectors, as explained above. In the case of a 2-dimensional manifold K_P is the Gaussian curvature at P . A *homogeneous* space, with maximal symmetry, will clearly have $K = \text{constant}$ over the space and (10.63) can be shown to imply (10.62). Let us see how this works for a 3-dimensional space. From (10.62), using Latin indices in accordance with our convention for a 3-dimensional space, the Ricci tensor is

$${}^{(3)}R_{ik} = g^{lm} R_{ilkm} = K(3g_{ik} - g_{ik}) = 2Kg_{ik}, \quad (10.65)$$

$$[{}^{(n)}R_{ik} = (n-1)Kg_{ik} \text{ in } n \text{ dimensions}] \quad (10.65a)$$

and the curvature scalar is

$${}^{(3)}R = g^{ik} {}^{(3)}R_{ik} = 6K = \frac{6}{a^2}, \quad (10.66)$$

$$[{}^{(n)}R = n(n-1)K \text{ in } n \text{ dimensions}] \quad (10.66a)$$

where in the last step we have used (10.57) for the 3-sphere. This result, (10.65), agrees with a direct calculation for S^3 (Problem 10.1).

We close this section by making some remarks about the geometry of S^3 , a contender for the space section of the Universe. Like the 2-sphere S^2 it is a *closed* space, one with no boundary. We write the line element in the form (see (10.51))

$$d\sigma^2 = \frac{dr^2}{1 - r^2/a^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (10.67)$$

S^3 is a homogeneous space and the origin may be chosen anywhere. The circumference of a circle is found by considering points distinguished only by their θ coordinate, so $dr = d\phi = 0$ and $d\sigma = r d\theta$, hence the circumference is $2\pi r$. Likewise the surface area of a 2-sphere in S^3 is

$$A = \int_0^\pi r^2 \sin\theta d\theta d\phi = 4\pi r^2.$$

The ‘radius’ of a circle, or sphere, is

$$\text{radius} = \int_0^r \left(1 - \frac{r'^2}{a^2}\right)^{-1/2} dr' = a \sin^{-1}(r/a) > r,$$

so

$$\frac{\text{circumference of circle}}{\text{radius of circle}} < 2\pi. \quad (10.68)$$

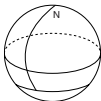


Fig. 10.5

Circles on the surface of a sphere. Measured from the north pole N the radius of the circle on the equator is the length of the line on the sphere joining them. A circle in the southern hemisphere has a larger radius but a smaller circumference than that at the equator.

This is also true of a circle S^1 inscribed on a sphere S^2 . There are other parallels with S^2 . To see these first write the line element (10.67) in polar coordinates (10.52):

$$(S^3): \quad d\sigma^2 = a^2[d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2)]. \quad (10.69)$$

The coordinate χ plays the role of distance; the distance from the origin is $a\chi$. The largest distance in the space is $a\pi$. There is an analogy on the 2-sphere, where

$$(S^2): \quad dl^2 = a^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (10.70)$$

Here θ plays a similar role: the distance from any origin is $a\theta$ and $a\pi$ is the largest distance on S^2 – that between the north and south poles.

The surface area of a 2-sphere in S^3 is clearly $4\pi a^2 \sin^2\chi$. As χ increases from 0, this reaches a maximum at $\chi = \pi/2$, at a distance of $a\pi/2$ and thereafter decreases with increasing distance, shrinking to a point as $\chi \rightarrow \pi$, a distance $a\pi$ away (the ‘opposite pole’). This is completely analogous to the description of circles on a 2-sphere (S^1 on S^2). Starting at, say, the north pole, the circumference of a circle initially increases with increasing distance away (‘radius’, but measured, of course, on the surface S^2 itself), but it reaches a maximum at the equator, and thereafter, when the circle is in the southern hemisphere, as its distance from the north pole (‘radius’) increases its circumference *decreases* (see Fig. 10.5), shrinking to a point at the south pole, the maximum distance away. Stepping up the dimensions by 1, then, to consider 2-spheres in S^3 , the observation is exactly that made above, that with increasing distance away, the area of the 2-sphere initially increases, but reaches a maximum beyond which it decreases with increasing radius (distance away) – we may say that bigger spheres ‘fit inside’ smaller spheres.

The volume of the space S^3 is

$$V = \int_0^{2\pi} \int_0^\pi \int_0^\pi a^3 \sin^2\chi \sin\theta \, d\chi \, d\theta \, d\phi = 2\pi^2 a^3;$$

a *finite* value. This space, like S^3 , has no boundary

$$\partial S^3 = 0; \quad (10.71)$$

it is a *closed* space. So, as pointed out by Landau and Lifshitz,⁶ Gauss's theorem implies that the total electric charge in the space must be zero, since (see (3.93), (3.95))

$$Q = \int_{S^3} \rho \, dV = \epsilon_0 \int_{S^3} \nabla \cdot \mathbf{E} \, dV = \epsilon_0 \int_{\partial S^3} \mathbf{E} \cdot \mathbf{n} \, d\Sigma = 0;$$

the total electric charge in a closed universe must be zero.

10.3 Hubble's law and the cosmological red-shift

The Robertson–Walker metric allows us to make various *kinematic* deductions, independent of the specific form of $S(t)$. The first of these is the Hubble expansion. Suppose that our Galaxy, comoving in the space-time continuum, is located at $r=0$, and another galaxy is at an arbitrary parameter r . Its proper distance L from us at cosmic time t is, from (10.62)

$$L = S(t) \int_0^r \frac{dr'}{\sqrt{1 - kr'^2}} = S(t)f(r), \quad (10.72)$$

with

$$f(r) = \begin{cases} \sin^{-1}(r) & (k = 1) \\ r & (k = 0) \\ \sinh^{-1}(r) & (k = -1) \end{cases} \quad (10.73)$$

so $L \propto S(t)$: the distance away changes with time (hardly surprisingly!). The *velocity* of recession is

$$v = \frac{dL}{dt} = \dot{S}(t) \int_0^r \frac{dr'}{\sqrt{1 - kr'^2}} = \frac{\dot{S}(t)}{S(t)} L, \quad (10.74)$$

hence

$$v \propto L \quad (10.75)$$

at any instant. This is Hubble's law $v = HL$ ((10.2)) with

$$H(t) = \frac{\dot{S}(t)}{S(t)}. \quad (10.76)$$

Note that Hubble's constant depends on time in general.

Now consider light reaching us, at $r=r_2$, having been emitted from a galaxy at $r=r_1$ – see Fig. 10.6. In particular consider successive crests of light, emitted at times t_1 and $t_1 + \Delta t_1$ and received at times t_2 and $t_2 + \Delta t_2$. Since $ds^2 = 0$ and the light is travelling radially we have

⁶ Landau & Lifshitz (1971), p. 375.

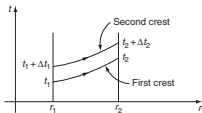


Fig. 10.6

Light emitted from a galaxy at $r = r_1$ reaches us at $r = r_2$.

$$0 = -c^2 dt^2 + S^2(t) \frac{dr^2}{1 - kr^2}, \quad (10.77)$$

so for the first crest of light

$$\int_{t_1}^{t_2} \frac{dt}{S(t)} = \frac{1}{c} \int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - kr^2}},$$

and for the second crest

$$\int_{t_1 + \Delta t_1}^{t_2 + \Delta t_2} \frac{dt}{S(t)} = \frac{1}{c} \int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - kr^2}};$$

hence

$$\int_{t_1 + \Delta t_1}^{t_2 + \Delta t_2} \frac{dt}{S(t)} = \int_{t_1}^{t_2} \frac{dt}{S(t)}.$$

Now Δt_1 and Δt_2 are very small so we may assume that $S(t)$ is constant over these intervals. Then, since

$$\int_{t_1 + \Delta t_1}^{t_2 + \Delta t_2} = \int_{t_1}^{t_2} + \int_{t_2}^{t_2 + \Delta t_2} - \int_{t_1}^{t_1 + \Delta t_1}$$

we find

$$\frac{\Delta t_1}{S(t_1)} = \frac{\Delta t_2}{S(t_2)} \Rightarrow \frac{\Delta t_1}{\Delta t_2} = \frac{S(t_1)}{S(t_2)}.$$

The emitted and observed wavelengths are

$$\lambda_e = c \Delta t_1, \quad \lambda_o = c \Delta t_2$$

so the *wavelength shift* is

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e} = \frac{S(t_1)}{S(t_2)} - 1. \quad (10.78)$$

If the Universe is *expanding*, $S(t_1) > S(t_2)$ so $z > 0$, resulting in a *red-shift*, as observed by Hubble. This is a *cosmological* red-shift and is conceptually distinct from a Doppler shift. In a Doppler shift both the source of the light and the observer are in the same inertial frame, and are moving relatively to each other. In the cosmological setting both the source and the observer are comoving; they are each in their own inertial frame. Their relative motion is due to the expansion of space. And, of course, both of these red-shifts are distinct from the gravitational red-shift.

It is worth making a remark at this point on the Steady State Theory. In this theory, since the Universe presents the same aspect at all times, Hubble's constant must be time independent, so (10.76) gives

$$\frac{dS}{dt} = HS, \quad S(t) = S(0) e^{Ht}; \quad (10.79)$$

the expansion of space is exponential, and $S(t)$ is never zero. Moreover, the curvature of the 3-space (10.67) is $6/a^2 = 6k$, so the curvature of the 3-space part of the Robertson-Walker metric (10.62) is $\frac{6k}{S^2(t)}$. In Steady State Theory this must be *constant* so we must have $k = 0$, and the steady state line element is

$$ds^2 = -c^2 dt^2 + e^{2Ht} [dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2)]. \quad (10.80)$$

Space is *flat* and the metric is determined on *kinematic grounds alone*. This metric is of interest not because the Steady State Theory is of interest but because it has the same form as the de Sitter metric, which we shall meet below.

10.4 Horizons

Let us suppose that the Universe 'began' at time t_B . In most models this will be a finite time in the past, which we may put equal to 0; in the steady state model it is equal to $-\infty$. Forgetting, temporarily, the expansion of the Universe, we may sketch our world-line, as in Fig. 10.7, from t_B to t_0 , which is 'now'. In this simple view of things, at t_B there were many galaxies - 'objects' or 'particles' - and it must be true that light from some of these galaxies is only now reaching us for the very first time. These objects are on a 'horizon', called an *object horizon* or *particle horizon*; they are the most distant objects we can now see. From (10.77) the (proper) distance to the particle horizon, r_{PH} , is given by

$$\int_{t_B}^{t_0} \frac{dt}{S(t)} = \frac{1}{c} \int_0^{r_{PH}} \frac{dr}{\sqrt{1 - kr^2}} = \frac{1}{c} \sin^{-1} r_{PH} \quad (k = 1) \quad (10.81)$$

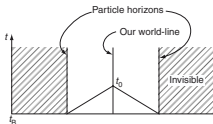


Fig. 10.7 Particle horizon: t_B is the 'beginning' of the Universe, t_0 is 'now'. In a non-expanding model light from sources further away than ct_0 has not yet reached us.

or

$$r_{\text{PH}} = \sin \left(c \int_{t_B}^{t_0} \frac{dt}{S(t)} \right) \quad (k = 1), \quad (10.82)$$

with analogous expressions for $k=0, -1$. Objects beyond r_{PH} cannot now be seen by us. Clearly r_{PH} increases with t_0 ; we shall be able to see more in the future than we can see now. In fact the horizon at r_{PH} is like an outgoing spherical wave centred on us. In addition, if there is now someone whom we are able to see for the first time, then 'they' must by symmetry be able to see us for the first time; an outgoing light-front leaving our world-line at t_B will now be reaching them, as also shown in Fig. 10.7.

In certain models we might find $r_{\text{PH}} = \infty$, and then the whole Universe can be seen by us, and there is no object horizon. In the $k=1$ case if

$$c \int_{t_B}^{t_0} \frac{dt}{S(t)} \geq \pi$$

then from (10.80) there is no object horizon, since π is the maximum distance in the S^3 model.

Now let us consider a related situation, but this time concerned with the *end* of the Universe, at t_E (which will be finite in some models, but infinite in others). We ask: what is the coordinate, r_{EH} , of the most distant event occurring *now* (at t_0) that we shall *ever* be able to see? This defines an *event horizon*. Light from the event horizon must reach us before, or as, the Universe ends, at t_E . So

$$\int_{t_0}^{t_E} \frac{dt}{S(t)} = \frac{1}{c} \int_0^{r_{\text{EH}}} \frac{dr}{\sqrt{1 - kr^2}}$$

or

$$r_{\text{EH}} = \sin \left(c \int_{t_0}^{t_E} \frac{dt}{S(t)} \right) \quad (k = 1), \quad (10.83)$$

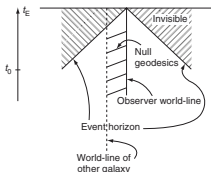


Fig. 10.8

Event horizon: t_E is the 'end' of the Universe, t_0 is 'now'. Light emitted from galaxies inside the event horizon will reach us before the Universe ends.

with analogous expressions for the cases $k = 0, -1$. Events occurring now beyond r_{EH} will never be seen by us – see Fig. 10.8. If $r_{EH} = \infty$, every event will at some time be seen by us (by an observer on our world-line). The event horizon is an inward-converging light cone reaching us at $t = t_E$. In Fig. 10.8 is drawn the world-line of another galaxy, with null geodesics, along which light will travel, from that galaxy to our own. It is clear that, as time goes on (as t_0 increases) light from this other galaxy will eventually cease to reach us before the Universe ends at $t = t_E$, so eventually every galaxy passes out of the event horizon of every other galaxy.

The above reasoning was based on the assumption that the Universe is not expanding, but we shall now show that concepts of horizons are basically unaffected by taking the expansion into account. Defining a new time parameter T by

$$dT = \frac{dt}{S(t)},$$

the R–W metric (10.62) becomes

$$ds^2 = S^2(t) \left(-c^2 dT^2 + \frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right), \quad (10.84)$$

which is *conformally related* to the non-expanding metric

$$ds^2 = -c^2 dT^2 + \frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2);$$

these line elements being obtained from one another by an overall conformal factor, as explained in Section 7.7. As was seen there, this conformal rescaling does not affect null geodesics (since $ds^2 = 0 \Leftrightarrow d\hat{s}^2 = 0$); it only has the effect of 'bringing in' points at infinity to the finite domain. Since the phenomena of event and particle horizons depend essentially on null geodesics, qualitative conclusions drawn from a static universe will also hold in an expanding one.

10.5 Luminosity-red-shift relation

The expansion of the Universe is governed by the function $S(t)$ and Hubble's constant $H(t)$ is, as we have seen, proportional to $\dot{S}(t)$. The double derivative $\ddot{S}(t)$ will clearly describe the *deceleration* of the expansion: on simple Newtonian grounds the Universe self-gravitates so we would expect the expansion to be slowing down and $\ddot{S}(t)$ to be non-zero – in fact, negative. The ‘deceleration parameter’ q , to be defined below, as well as $H(t)$, are quantities that play key roles in cosmology, since they may be derived from theoretical cosmological models but they may also be found by exploring relations between various *observables*. One of these is the one between luminosity and red-shift, which we now discuss.

Let us calculate the light flux received from a distant source. Suppose the light is emitted at (cosmic) time t_e and radial coordinate r_e , and is just now reaching us, at time t_0 and coordinate r_0 . This is shown in Fig. 10.9 where we have chosen to put $r_0 = 0$; r decreases as t increases along the light path. Considering, on the other hand, the *source* to be at the centre of a sphere, the light is now crossing the surface of this sphere, whose area is

$$4\pi r_e^2 S^2(t_0).$$

Suppose the source has absolute luminosity L ; this is the amount of energy emitted per unit time (i.e. power). The total power *received* per unit area (for example through a telescope) is not simply L divided by the area above, since there are two red-shift factors to be taken into account:

- (i) Each *photon* is red-shifted. From (10.78) the ratio of observed and emitted wavelengths and frequencies is

$$\frac{\lambda_0}{\lambda_e} = z + 1 = \frac{S(t_0)}{S(t_e)} = \frac{v_e}{v_0} = \frac{E_e}{E_0},$$

so the energy of each photon is decreased by $(1 + z)$.

- (ii) The *rate of arrival* of photons is decreased by the same factor. Two photons emitted within a time interval δt_e will arrive within the time interval

$$\delta t_0 \cdot \frac{S(t_0)}{S(t_e)} = \delta t_e (1 + z).$$

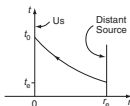


Fig. 10.9

Light emitted from a source at r_e at time t_e reaches us, at a position $r_0 (= 0 \text{ here})$ at time t_0 .

Hence the power received per unit area, which is the *apparent luminosity* l , is⁷

$$l = \frac{L}{4\pi r_e^2 S^2(t_0)} \cdot \frac{1}{(1+z)^2} = \frac{LS^2(t_e)}{4\pi r_e^2 S^4(t_0)}. \quad (10.85)$$

A parameter clearly related to apparent luminosity is the *luminosity distance* d_L defined by

$$l = \frac{L}{4\pi d_L^2}. \quad (10.86)$$

We now turn to *observables*. Besides Hubble's constant

$$H(t) = \frac{\dot{S}(t)}{S(t)}, \quad (10.87)$$

we also define the *deceleration parameter* $q(t)$,

$$q(t) = -\frac{\ddot{S}(t)}{\dot{S}(t)^2}. \quad (10.88)$$

Expanding about $t = t_0$ and with

$$H_0 = H(t_0), \quad q_0 = q(t_0) \quad (10.89)$$

we have

$$\begin{aligned} S(t) &= S(t_0) + (t - t_0)\dot{S}(t_0) + \frac{1}{2}(t - t_0)^2\ddot{S}(t_0) + \dots \\ &= S(t_0) \left[1 + (t - t_0)H_0 - \frac{1}{2}(t - t_0)^2 q_0 H_0^2 + \dots \right]. \end{aligned} \quad (10.90)$$

We now want to express $(t - t_0)$ in terms of the red-shift z , an observable quantity. From (10.78), and using (10.90),

$$\begin{aligned} z &= S(t_0) S^{-1}(t_e) - 1 = [1 + (t_e - t_0)H_0 - \frac{1}{2}(t_e - t_0)^2 q_0 H_0^2 + \dots]^{-1} - 1 \\ &= H_0(t_0 - t_e) + \left(1 + \frac{q_0}{2}\right) H_0^2(t_0 - t_e)^2 + \dots \end{aligned}$$

Inverting this gives

$$t_0 - t_e = \frac{z}{H_0} - \left(1 + \frac{q_0}{2}\right) H_0 \left(\frac{z}{H_0}\right)^2 = \frac{1}{H_0} \left[z - \left(1 + \frac{q_0}{2}\right) z^2 \right]. \quad (10.92)$$

Now let us find an expression for r_e . From (10.62), for light travelling radially we have

$$\int_{t_e}^{t_0} \frac{dt}{S(t)} = \int_0^{r_e} [1 - kr^2]^{-1/2} dr. \quad (10.93)$$

⁷ This argument, invoking photons, clearly relies on quantum theory, which is hardly ideal in the context of General Relativity. For an argument for inclusion of the factor $(1+z)^{-2}$, which is based on GR alone, see Robertson (1938).

The left hand side of this is, from (10.90)

$$\begin{aligned} \text{lhs} &= S^{-1}(t_0) \int_{t_c}^{t_0} [1 - H_0(t - t_0) + H_0^2(t - t_0)^2 + \frac{q_0}{2} H_0^2(t - t_0)^2 + \dots] dt \\ &= S^{-1}(t_0) [(t_0 - t_c) + 1/2 H_0(t_0 - t_c)^2 + \dots], \end{aligned}$$

while the right hand side is

$$\text{rhs} = r_c + O(r_c^3),$$

so to lowest order in r_c and using (10.93)

$$\begin{aligned} r_c &= \frac{1}{S(t_0)} \left[(t_0 - t_c) + \frac{H_0}{2} (t_0 - t_c)^2 + \dots \right] = \frac{1}{S(t_0) H_0} \left[z - \left(1 + \frac{q_0}{2}\right) z^2 + \frac{z^3}{2} \right] \\ &= \frac{1}{S(t_0) H_0} [z - 1/2 (1 + q_0) z^2]. \end{aligned} \quad (10.94)$$

Now substituting successively (10.94), (10.91) and (10.93) into (10.85) gives

$$\begin{aligned} l &= \frac{L}{4\pi} \cdot \frac{S^2(t_c)}{S^2(t_0)} \cdot \frac{H_0^2}{z^2} [1 - 1/2(1 + q_0)z]^{-2} \\ &= \frac{LH_0^2}{4\pi z^2} [1 + (t_c - t_0)H_0 + \dots]^2 [1 + (1 + q_0)z] \\ &= \frac{LH_0^2}{4\pi z^2} (1 - 2z) [1 + (1 + q_0)z] \\ &= \frac{LH_0^2}{4\pi z^2} [1 + (q_0 - 1)z]. \end{aligned} \quad (10.95)$$

This is the *luminosity-red-shift relation*. An equivalent relation is the *distance-red-shift relation*, which follows from (10.95) and (10.86):

$$d_L = \frac{1}{H_0} [z - 1/2(q_0 - 1)z^2]. \quad (10.96)$$

From these relations between observable quantities it is, in principle, possible to find H_0 and q_0 . Hubble's constant is the easier parameter to find, being simply related to the gradient of the d_L - z graph for small z . Recent estimates give⁸

$$H_0 = (55 - 85) \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (10.97)$$

or

$$H_0 = h \cdot 100 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad h = (0.55 - 0.85). \quad (10.98)$$

The determination of q_0 has, however, seen some rather dramatic developments over the last 10 years. Before about 1998, again working in the area of small red-shifts (z up to about 0.4

⁸ Freedman (1997).

or 0.5), q_0 was reckoned to be between 0 and 1; a fairly large uncertainty, but *positive*, corresponding to a *decelerating* universe, understandable through Newtonian intuition.

More recently, however, the data have improved and the reason for this is related to improved estimates of distance. To know the luminosity distance d_L we have to know the absolute luminosity of the source L – see (10.86). In general this gives rise to a problem: if a star is faint it could be because it is *intrinsically* faint, or because it is brighter but further away. How do we know which? We rely on so-called ‘standard candles’, sources whose absolute magnitudes are known. The classic example is a Cepheid variable. These are stars whose luminosity varies periodically, but with a period dependent on the luminosity. Using Cepheid variables enables us to find d_L , and measuring the red-shift earns the star a place on the d_L – z plot. Beyond a particular distance, however, Cepheid variable stars become too faint. Their role as standard candles, has, however, now been taken over by *type Ia supernovae*. These are binary white dwarf systems, whose heavier partner attracts material (mass) from the lighter partner until its mass exceeds the Chandrasekhar limit (see Section 7.3), resulting in an explosion leading to a supernova – bright, and visible from far away! Crucially, the luminosity of an exploding white dwarf is a fairly good standard candle so the distance of these supernovae can then be estimated and they enable more data to appear on the d_L – z plot, with higher values of d_L and z , so that in particular the *non-linear* term in (10.96) can be found with some accuracy. And, surprisingly, q_0 turns out to be *negative* – the expansion of the Universe is actually *accelerating*, not slowing down. Newtonian intuition fails here, but it turns out that this scenario is not new in cosmology. Einstein introduced his famous cosmological constant Λ to solve a problem that has gone away, but Λ corresponds exactly to an accelerating expansion. We shall return to this topic below.

10.6 Dynamical equations of cosmology

The Robertson–Walker metric (10.62)

$$ds^2 = -c^2 dt^2 + S^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right) \quad (10.100)$$

contains the ‘free’ functions $S(t)$ and k , describing the expansion and the curvature of space. The Einstein field equations (5.24)

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^2} T_{\mu\nu} \quad (10.101)$$

will yield equations for $S(t)$ and k so enabling us to see what sort of universe General Relativity permits (provided we have an expression for the energy-momentum tensor).

The quickest way to find the left hand side of (10.101) is to use differential forms. The metric (10.100) may be written

$$ds^2 = -(\theta^0)^2 + (\theta^1)^2 + (\theta^2)^2 + (\theta^3)^2 \quad (10.102)$$

which, with basis 1-forms

$$\theta^0 = c \, dt, \quad \theta^1 = Su^{-1} \, dr, \quad \theta^2 = Sr \, d\theta, \quad \theta^3 = Sr \sin \theta \, d\phi, \quad (10.103)$$

where

$$u(r) = \sqrt{1 - kr^2} \quad (10.104)$$

is an *orthonormal* (anholonomic) basis with metric tensor

$$g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1). \quad (10.105)$$

The exterior derivatives of the basis forms are

$$d\theta^0 = 0, \quad (10.106a)$$

$$d\theta^1 = \dot{S}u^{-1} \, dt \wedge dr = \frac{\dot{S}}{cS} \theta^0 \wedge \theta^1, \quad (10.106b)$$

$$d\theta^2 = \dot{S}r \, dt \wedge d\theta + S \, dr \wedge d\theta = \frac{\dot{S}}{cS} \theta^0 \wedge \theta^2 + \frac{u}{Sr} \theta^1 \wedge \theta^2, \quad (10.106c)$$

$$\begin{aligned} d\theta^3 &= \dot{S}r \sin \theta \, dt \wedge d\phi + S \sin \theta \, dr \wedge d\phi + Sr \cos \theta \, d\theta \wedge d\phi \\ &= \frac{\dot{S}}{cS} \theta^0 \wedge \theta^3 + \frac{u}{Sr} \theta^1 \wedge \theta^3 + \frac{\cot \theta}{Sr} \theta^2 \wedge \theta^3. \end{aligned} \quad (10.106d)$$

The structure equations (3.182)

$$d\theta^\mu + \omega^\mu{}_\kappa \wedge \theta^\kappa = 0 \quad (10.107)$$

give, for equation (10.106a),

$$\omega^0{}_1 \wedge \theta^1 + \omega^0{}_2 \wedge \theta^2 + \omega^0{}_3 \wedge \theta^3 = 0, \quad (10.108)$$

where we have used the fact that

$$\omega_{\mu\nu} = -\omega_{\nu\mu}, \quad (10.109)$$

which follows from (10.105) and (3.191). Equations (10.106b) and (10.107) give

$$-\omega^1{}_0 \wedge \theta^0 - \omega^1{}_2 \wedge \theta^2 - \omega^1{}_3 \wedge \theta^3 = \frac{\dot{S}}{cS} \theta^0 \wedge \theta^1,$$

which implies that

$$\omega^1{}_0 = \frac{\dot{S}}{cS} \theta^1 = \frac{\dot{S}}{c} u^{-1} \, dr. \quad (10.110)$$

Next, (10.106c) gives

$$-\omega^2{}_0 \wedge \theta^0 - \omega^2{}_1 \wedge \theta^1 - \omega^2{}_3 \wedge \theta^3 = \frac{\dot{S}}{cS} \theta^0 \wedge \theta^2 + \frac{u}{Sr} \theta^1 \wedge \theta^2,$$

which implies that

$$\omega^2_0 = \frac{\dot{S}}{cS} \theta^2 = \frac{r\dot{S}}{c} d\theta, \quad \omega^2_1 = \frac{u}{Sr} \theta^2 = u d\theta, \quad (10.111)$$

and finally (10.106d) and (10.108) give

$$-\omega^3_0 \wedge \theta^0 - \omega^3_1 \wedge \theta^1 - \omega^3_2 \wedge \theta^2 = \frac{\dot{S}}{cS} \theta^0 \wedge \theta^3 + \frac{u}{Sr} \theta^1 \wedge \theta^3 + \frac{\cot \theta}{Sr} \theta^2 \wedge \theta^3$$

and hence

$$\begin{aligned} \omega^3_0 &= \frac{\dot{S}}{cS} \theta^3 = \frac{\dot{S}}{c} r \sin \theta d\phi, \\ \omega^3_1 &= \frac{u}{Sr} \theta^3 = u \sin \theta d\phi, \\ \omega^3_2 &= \frac{\cot \theta}{Sr} \theta^3 = \cos \theta d\phi. \end{aligned} \quad (10.112)$$

Note that equations (10.110)–(10.112) satisfy (10.108)–in fact each term separately vanishes in this equation.

Having found the connection 1-forms we now calculate the curvature 2-forms from (4.45)

$$\Omega^\mu_\nu = d\omega^\mu_\nu + \omega^\mu_\lambda \wedge \omega^\lambda_\nu. \quad (10.113)$$

We have

$$\Omega^1_0 = d\omega^1_0 + \omega^1_2 \wedge \omega^2_0 + \omega^1_3 \wedge \omega^3_0.$$

The second and third terms vanish and

$$\Omega^1_0 = \frac{\ddot{S}}{c} u^{-1} dt \wedge dr = \frac{\ddot{S}}{c^2 S} \theta^0 \wedge \theta^1. \quad (10.114)$$

The Riemann curvature tensor follows from (4.39)

$$\Omega^\mu_\nu = 1/2 R^\mu_{\nu\rho\sigma} \theta^\rho \wedge \theta^\sigma, \quad (10.115)$$

hence

$$R^1_{001} = \frac{\ddot{S}}{c^2 S} = -R^1_{010} \quad (10.116)$$

and all other $R^1_{0\rho\sigma}$ vanish. Similarly we have

$$\Omega^2_0 = d\omega^2_0 + \omega^2_1 \wedge \omega^1_0 + \omega^2_3 \wedge \omega^3_0 = \frac{r\ddot{S}}{c} dt \wedge d\theta = \frac{\ddot{S}}{c^2 S} \theta^0 \wedge \theta^2 \quad (10.117)$$

and (10.115) gives

$$R^2_{002} = \frac{\ddot{S}}{c^2 S} = -R^2_{020}. \quad (10.118)$$

The same equation holds for R^3_{003} so we have

$$R^1_{010} = R^2_{020} = R^3_{030} = -\frac{\ddot{S}}{c^2 S}. \quad (10.119)$$

The remaining components of the Riemann tensor are found using, for example

$$\Omega^2_1 = d\omega^2_1 + \omega^2_0 \wedge \omega^0_1 + \omega^2_3 \wedge \omega^3_1 = \left(\frac{u u'}{S^2 r} - \frac{\dot{S}^2}{c^2 S^2} \right) \theta^1 \wedge \theta^2, \quad (10.120)$$

from which

$$R^2_{121} = \frac{1}{S^2} \left(k + \frac{\dot{S}^2}{c^2} \right). \quad (10.121)$$

Similarly we have

$$R^3_{131} = R^3_{232} = R^2_{121} = \frac{1}{S^2} \left(k + \frac{\dot{S}^2}{c^2} \right). \quad (10.122)$$

From these equations the components of the Ricci tensor are easily found. For example

$$\begin{aligned} R_{00} &= R^{\mu}_{0\mu 0} = R^1_{010} + R^2_{020} + R^3_{030} = -3 \frac{\ddot{S}}{c^2 S}, \\ R_{11} &= R^0_{101} + R^2_{121} + R^3_{131} = \frac{\ddot{S}}{c^2 S} + \frac{2}{S^2} \left(k + \frac{\dot{S}^2}{c^2} \right) = R_{22} = R_{33}. \end{aligned} \quad (10.124)$$

The curvature scalar may then be found (recall the metric (10.105))

$$R = \eta^{00} R_{00} + \eta^{11} R_{11} + \eta^{22} R_{22} + \eta^{33} R_{33} = -R_{00} + 3R_{11} = 6 \frac{\ddot{S}}{c^2 S} + \frac{6}{S^2} \left(k + \frac{\dot{S}^2}{c^2} \right). \quad (10.125)$$

In addition it is found that

$$R_{12} = R_{13} = R_{23} = R_{01} = R_{02} = R_{03} = 0. \quad (10.126)$$

For the energy-momentum tensor we adopt the model of the *perfect fluid*. From (7.8) this is

$$T_{\mu\nu} = \frac{P}{c^2} g_{\mu\nu} + \left(\rho + \frac{P}{c^2} \right) u_{\mu} u_{\nu}. \quad (10.127)$$

In the comoving frame $u^0 = u_0 = 1$, $u^i = u_i = 0$ and $g_{\mu\nu} = \eta_{\mu\nu}$, so

$$T_{00} = \rho, \quad T_{11} = T_{22} = T_{33} = \frac{P}{c^2}, \quad T_{ij} = 0 \ (i \neq j), \quad T_{0i} = 0. \quad (10.128)$$

We are now in a position to write down the field equations (10.101). The ‘time-time’ component

$$R_{00} - \frac{1}{2} \eta_{00} R = \frac{8\pi G}{c^2} T_{00}$$

(recall the metric (10.105)!) gives, with (10.123), (10.125) and (10.128)

$$\dot{S}^2 + kc^2 = \frac{8\pi G}{3} \rho S^2 \quad (10.129)$$

and the 'space-space' component

$$R_{11} - \frac{1}{2}\eta_{11}R = \frac{8\pi G}{c^2} T_{11}$$

gives

$$2S\ddot{S} + \dot{S}^2 + kc^2 = -\frac{8\pi G}{c^2} p S^2. \quad (10.130)$$

Equation (10.129) is called the *Friedmann equation*,⁹ though more loosely the models obtained by using particular equations of state are also called Friedmann models – see the next section.

There is a compatibility condition linking these last two equations. Differentiating (10.129) and substituting (for $\ddot{S}$) into (10.130) gives, after a bit of algebra

$$\dot{\rho} + 3\left(\rho + \frac{p}{c^2}\right)\frac{\dot{S}}{S} = 0. \quad (10.131)$$

This is the compatibility condition for (10.129) and (10.130). In what follows it is convenient to treat (10.129) and (10.131) as the fundamental equations. They are differential equations and will be solved below. Firstly, however, it is interesting to consider a Newtonian interpretation of them.

10.6.1 Newtonian interpretation

Consider an expanding Newtonian universe with a centre, and a galaxy of mass m at a distance $r = r_0 S(t)$ from the centre (see Fig. 10.10). Its kinetic and potential energies are

$$\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2}mr_0^2\dot{S}^2, \quad \text{PE} = -m\left(\frac{4\pi}{3}r^3\rho\right)\frac{G}{r} = -\frac{4\pi G}{3}m\rho r^2,$$

and conservation of energy gives



Fig. 10.10

A Newtonian universe. A galaxy of mass m is at a distance r from the centre.

⁹ Friedmann (1922).

$$\begin{aligned} \frac{1}{2} m r_0^2 \dot{S}^2 - \frac{4\pi G}{3} m \rho r_0^2 S^2 &= E, \\ \dot{S}^2 - \frac{8\pi G}{3} \rho S^2 &= \frac{2mE}{r_0^2} = \text{const}, \end{aligned}$$

which is (10.129) with $kc^2 = -\frac{2mE}{r_0^2}$.

Furthermore, since the expansion of the Universe must be adiabatic, conservation of entropy gives

$$dU = -p dV$$

where U is the internal energy $= \rho c^2 V$. Hence

$$\frac{\partial}{\partial t} \left(\frac{4\pi}{3} r^3 \rho c^2 \right) = -p \frac{\partial}{\partial t} \left(\frac{4\pi}{3} r^3 \right)$$

or

$$\frac{\partial \rho}{\partial t} + 3 \frac{\dot{S}}{S} \left(\rho + \frac{p}{c^2} \right) = 0,$$

which is (10.131). Equations (10.129) and (10.131) therefore express, in a Newtonian interpretation, conservation of energy and of entropy.

10.6.2 Critical density

Let us now investigate the consequences of Equations (10.129) and (10.131). Subtracting (10.129) from (10.130) gives

$$\ddot{S} = -\frac{4\pi G}{3} \left(\rho + 3 \frac{p}{c^2} \right) S, \quad (10.132)$$

showing that $\ddot{S}$ is *always negative*, so $\dot{S}$ is always decreasing in time – the Universe was expanding faster in the past than it is now. This is of course precisely what our Newtonian intuition would lead us to expect, and has already been mentioned earlier in this chapter. We now see, though, that the slowing expansion is actually a consequence of General Relativity. An expanding universe means that S was smaller in the past than it is now, and there must have been a time when it was ‘zero’ (zero, that is, in this classical model, in which masses are point masses and quantum effects are ignored). This was the initial singularity that marked the ‘birth’ of our Universe. Putting $t = 0$ at $S = 0$ and denoting, as before, the present age of the Universe by t_0 we see, by consulting Fig. 10.11, that since the gradient of the tangent at $t = t_0$ is $\dot{S}_0$ then from (10.76) the intercept on the time axis is H_0^{-1} , which must be *greater* than the age of the Universe. With $h = 0.55$ in (10.98) we have

$$t_0 < 1.8 \times 10^{10} \text{ years}$$

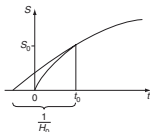


Fig. 10.11 A decelerating universe. Its age is less than the inverse of (the present value of) Hubble's constant.

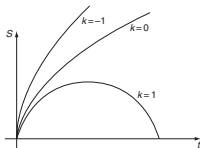


Fig. 10.12 Three types of cosmological model: $k = 1$ – expanding phase comes to an end and is followed by a contracting phase; $k = 0$ – expansion ceases when the Universe is 'infinitely dilute'; $k = -1$ – expansion never ceases – time without end.

as an upper limit. This is consistent with independent estimates of the age of the Earth (about 4.5×10^9 years) and of one of the oldest bright cluster galaxies¹⁰

$$t_{\text{BCG}} = 13.4^{+1.4}_{-1.0} \times 10^9 \text{ years.}$$

From (10.129), if $k = 0$ or -1 $\dot{S}^2$ is never zero, so the Universe expands for ever. If $k = 1$, however, the expansion will stop ($\dot{S} = 0$) when ρS^2 reaches the value $3kc^2/8\pi G$. Thereafter, since $\ddot{S} < 0$, $\dot{S}$ will become negative and the Universe will start to contract. These alternative scenarios are shown in Fig. 10.12. The fate of the Universe, then, depends on k , the sign of the spatial curvature. What is the value of k ?

Let us specialise (10.129) to the present moment ($t = t_0$); then

$$kc^2 = \frac{8\pi G}{3} \rho_0 S_0^2 - \dot{S}_0^2 = \frac{8\pi G}{3} S_0^2 \left(\rho_0 - \frac{3H_0^2}{8\pi G} \right).$$

The last term in brackets, which is clearly a density, is called the *critical density* ρ_c ,

$$\rho_c = \frac{3H_0^2}{8\pi G}. \quad (10.133)$$

¹⁰ Ferreras *et al.* (2001).

With $H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$\rho_c = 0.97 \times 10^{-26} \text{ kg m}^{-3} \quad (10.134)$$

or more generally

$$\rho_c = 1.88h^2 \times 10^{-26} \text{ kg m}^{-3}. \quad (10.135)$$

This density is important for it determines the sign of k : if $\rho > \rho_c$, k is positive and the Universe will eventually stop expanding and recontract, but if $\rho < \rho_c$, it will expand for ever – there will not be enough matter in it to halt the collapse. There is thus a connection between the curvature of space and the fate of the Universe; but note that these conclusions are limited to the case of a decelerating universe.

What is the *actual* density of matter (and radiation) in the Universe? The average density of *luminous* (or *baryonic*) matter – stars and galaxies – is of the order

$$\rho_b \approx 10^{-28} \text{ kg m}^{-3}, \quad (10.136)$$

approximately one proton in every 10 cubic metres. This is only a small percentage of the critical density (10.134). There is, however, very good evidence that there is a large amount of mass in the Universe which is *not* luminous; this is called *dark matter*. Dark matter is believed to exist, for example, in galactic haloes – clouds of gas rotating around a galactic nucleus. Suppose the velocity of the gas a distance r from the centre is $v(r)$ and the total mass out to a distance r is $M(r)$. Then from Newtonian mechanics we would expect that

$$\frac{GM(r)}{r^2} = \frac{v^2(r)}{r}.$$

For gas well outside the visible part of a galaxy we would expect that $M(r)$ is roughly constant (independent of r), and we would then have $v(r) \sim r^{-1/2}$. Now the velocity can be found by measuring the Doppler shift of the 21 cm hydrogen line, and what is found is that that $v(r)$ is more or less *constant*, so $M(r)$ is not constant, but grows proportionately to r , $M(r) \sim r$. Hence there must be *invisible* matter in the galaxy, with, as it turns out, a total density much greater than that of the luminous matter. The composition of dark matter is not known, and there is speculation that at least some of it is *non-baryonic* – not made of protons and neutrons, but perhaps other ‘exotic’ particles. This is an important area of modern cosmology but will not be pursued further here.

A useful parameter for discussing the contributions to the energy density of material in the Universe is

$$\Omega = \frac{\rho}{\rho_c}. \quad (10.137)$$

Recent estimates for the baryonic and matter contributions (and note that, in this definition, ‘matter’ includes dark matter *and* luminous matter) are¹¹

$$\Omega_b = 0.044 \pm 0.004, \quad (10.138)$$

¹¹ Spergel *et al.* (2003).

$$\Omega_m = 0.27 \pm 0.04. \quad (10.139)$$

As is discussed below (and as the reader is doubtless already aware) there is also thermal radiation in the Universe, at a temperature $T = 2.72$ K. This gives a radiation energy density

$$\rho_r c^2 = \frac{4\sigma}{c} T^4 = 4.16 \times 10^{-14} \text{ J m}^{-3}$$

in energy units, or in mass units

$$\rho_r = 4.62 \times 10^{-31} \text{ kg m}^{-3}, \quad (10.140)$$

giving, for the cosmic thermal radiation,

$$\Omega_r = 4.8 \times 10^{-5}. \quad (10.141)$$

10.7 Friedmann models and the cosmological constant

Let us make some observations about the dynamical equations (10.129) to (10.131) (or (10.132)) above:

$$\dot{S}^2 + kc^2 = \frac{8\pi G}{3} \rho S^2, \quad (10.129)$$

$$2S\ddot{S} + \dot{S}^2 + kc^2 = -\frac{8\pi G}{c^2} p S^2, \quad (10.130)$$

$$\dot{\rho} + 3\left(\rho + \frac{p}{c^2}\right)\frac{\dot{S}}{S} = 0. \quad (10.131)$$

These represent, as we have seen, only two independent equations for the three unknowns, ρ , p and S . A complete solution is possible if one or more relations between these quantities – for example an equation of state, relating ρ and p – is known. Models constructed in this way are called *Friedmann models*, after the Russian mathematician and physicist. There are two special cases of interest: dust, modelling ordinary matter, and radiation.

The dust model is a *zero pressure* model. We put $p = 0$ in (10.132), giving

$$\ddot{S} = -\frac{4\pi G}{3} \rho S. \quad (10.142)$$

Differentiating (10.129) gives

$$2\dot{S}\ddot{S} = \frac{8\pi G}{3} (2\rho S\dot{S} + S^2\dot{\rho}), \quad (10.143)$$

which, on substituting (10.142), gives

$$[\text{dust}] \quad \frac{\dot{\rho}}{\rho} = -3\frac{\dot{S}}{S} \Rightarrow \rho S^3 = \text{const} = \rho_0 S_0^3. \quad (10.144)$$

This is easy to understand: it is simply the consequence of conservation of mass for non-relativistic matter in an expanding space; $m = \rho V = \text{const}$. This describes a *matter-dominated universe*. Hardly surprisingly, it is a universe in which q , the deceleration parameter, is positive, since from (10.142) we have, for the present value of q

$$q_0 = -\frac{\ddot{S}_0 S_0}{\dot{S}_0^2} = \frac{4\pi G}{3} \rho_0 \frac{S_0^2}{\dot{S}_0^2} = \frac{4\pi G}{3} \frac{\rho_0}{H_0^2} > 0.$$

The second case of interest is the *radiation gas model*, in which¹²

$$\rho = \frac{3p}{c^2}. \quad (10.145)$$

Then (10.132) gives

$$\ddot{S} = -\frac{8\pi G}{3} \rho S.$$

Substituting this in (10.143) gives

$$[\text{radiation}] \quad \frac{\dot{\rho}}{\rho} = -4 \frac{\dot{S}}{S} \Rightarrow \rho S^4 = \text{const} = \rho_0 S_0^4. \quad (10.146)$$

This is to be compared with (10.144) for (non-relativistic) matter. To gain an understanding of the extra power of S in the radiation case we could argue that if a box of photons, of volume V , is expanded, the volume of the box increases by a factor S^3 , but in addition the energy of each photon $E = h\nu = ch/\lambda$ decreases by a factor $1/S$, since $\lambda \rightarrow S\lambda$ under an expansion of space, so the energy *density* decreases by $1/S^4$.

In a universe, therefore, containing matter *and* radiation (though decoupled), their respective densities will behave, under expansion, as

$$\rho_m \propto \frac{1}{S^3}, \quad \rho_r \propto \frac{1}{S^4}, \quad \frac{\rho_r}{\rho_m} \propto \frac{1}{S(t)}. \quad (10.147)$$

This is sketched in Fig. 10.13 (notionally a logarithmic plot). If there is *any* radiation in the Universe at the present time, it must have dominated at early times. The *present* densities of matter and radiation are given by (10.139) and (10.141), so the present Universe is

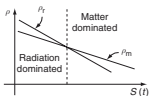


Fig. 10.13

The density of radiation decreases faster than the density of matter in an expanding universe.

¹² See for example Feynman *et al.* (1963), Section 39-3.

obviously matter-dominated, but at early enough times it must have been radiation-dominated. This assumption is a key ingredient in studying the physics of the early Universe.

Next, let us consider the question of *static solutions*, thought to be relevant before Hubble's discovery that the Universe is expanding. In a static universe $\dot{S} = \ddot{S} = 0$ and (10.129) and (10.130) then give

$$\rho = -\frac{3p}{c^2}; \quad (10.148)$$

the Universe is occupied by a fluid with negative pressure (if positive energy density). Dismissing this as unphysical we may conclude that the equations have *no static solutions*. For Einstein in 1917 this was a problem, which he addressed by introducing an extra term into the field equations,¹³ so that they now read (in modern notation – Einstein used λ rather than Λ)

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R - g_{\mu\nu}\Lambda = \frac{8\pi G}{c^2}T_{\mu\nu}, \quad (10.149)$$

and Λ is called the *cosmological constant*. Equations (10.129) and (10.130) then become

$$\dot{S}^2 + kc^2 = \frac{8\pi G}{3}\rho S^2 + \frac{c^2}{3}\Lambda S^2, \quad (10.150)$$

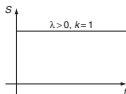
$$2S\ddot{S} + \dot{S}^2 + kc^2 = -\frac{8\pi G}{c^2}pS^2 + c^2\Lambda S^2. \quad (10.151)$$

Equation (10.131) is unchanged.

A static universe ($\dot{S} = \ddot{S} = 0$) with $p = 0$ (dust model) now follows if

$$k = \Lambda S^2, \quad \Lambda = \frac{4\pi G}{c^2}\rho. \quad (10.152)$$

This is a closed, static universe, the so-called *Einstein universe*, shown in Fig. 10.14. It is clear that a positive Λ must represent a 'repulsive force' to balance the gravitational attraction of matter. To see this, consider an empty flat universe, $\rho = k = 0$. From (10.150)



Einstein universe.

¹³ Einstein (1917).

Fig. 10.14

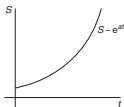


Fig. 10.15

de Sitter universe.

$$\ddot{S}^2 = \frac{c^2}{3} \Lambda S^2, \quad S(t) = S(0) \exp \left\{ \left(\frac{c^2 \Lambda}{3} \right)^{1/2} t \right\}. \quad (10.153)$$

This is the *de Sitter universe*, shown in Fig. 10.15. It describes an empty but expanding space. The space expands by ‘cosmic repulsion’. This is similar to the Steady State Theory, except that here $\rho = 0$. In the *de Sitter universe* the deceleration parameter is negative

$$q = -\frac{\ddot{S}S}{\dot{S}^2} = -1,$$

indicating an accelerating expansion, as observed in our Universe. It is clear then, that this feature of the Universe will be accounted for, in the language of General Relativity, by a positive cosmological constant Λ .

The introduction of the cosmological constant by Einstein, however, was not entirely trouble-free. If we put $\dot{S} = 0$, $k = 0$ and $p = 0$ in (10.151) we find, using (10.152)

$$\ddot{S} = \frac{c^2 \Lambda}{2} S = -4\pi\rho S,$$

hence $\ddot{S} < 0$ and the solution is *unstable*. This, together with Hubble’s later discovery that the Universe is expanding after all, led Einstein to abandon the Λ term, calling its introduction the ‘greatest blunder’ of his scientific life. As seen above, though, the fact that the expansion of the Universe is now seen to be accelerating indicates that there is, after all, a place for Λ , so Einstein’s blunder might turn out to be not such a bad idea after all!

Introducing Λ into the picture, however, considerably complicates matters; there are many types of solutions to Equations (10.150) and (10.151), depending on the sign and magnitude of k and Λ . Rather than attempt a general survey we shall consider simply a particular case, which however does seem to stand a reasonable chance of being correct. This is the case in which $k = 0$; the Universe is *flat*. There is in fact evidence for this from the now rather detailed observations of the cosmic background radiation, a topic to be discussed in the next section. The relevant observations are of the *anisotropy* of this radiation. The radiation reaching us now was last scattered some hundreds of thousands of years after the beginning of the universal expansion; and already at that time there were *inhomogeneities* in the cosmic dust, which would in the course of time result in the condensation of dust into galaxies. These inhomogeneities depend on the parameters k and Λ , which govern the dynamics of the evolution – and at the same time can in effect be measured by analysing the

inhomogeneities in the scattered radiation. The analysis of these is a rather complicated matter, but Hartle concludes that the evidence, as of 2003, is ‘consistent with’ a flat universe.¹⁴ Making this assumption and writing (10.150) in the form

$$kc^2 = \frac{8\pi G}{3} S_0^2 \left(\rho_0 + \frac{c^2 \Lambda}{8\pi G} - \frac{3H_0^2}{8\pi G} \right),$$

where the subscript 0 refers to the present time, then putting $k=0$ and $\rho = \rho_r + \rho_m$ we have, with (10.133),

$$\rho_r + \rho_m + \frac{c^2 \Lambda}{8\pi G} - \rho_c = 0.$$

We write this as (see (10.137))

$$\Omega_r + \Omega_m + \Omega_\Lambda = 1 \quad (10.154)$$

where

$$\Omega_\Lambda = \frac{c^2 \Lambda}{8\pi G} \cdot \frac{1}{\rho_c} = \frac{c^2 \Lambda}{3H_0^2} \quad (10.155)$$

The term Ω_Λ may be said to represent the contribution of the cosmological constant term to the energy density of the Universe. With Ω_r and Ω_m taking on the values (10.140) and (10.139) it is clear that Ω_Λ provides the dominant contribution (≈ 0.7) to the dynamics of the Universe. The physical nature of this contribution is still unclear, but it goes by the name of *dark energy*.

10.8 Cosmic background radiation

The discovery, in 1965, that the Universe contains thermal radiation, was a landmark in twentieth-century cosmology, equal in importance to Hubble’s discovery that the Universe is expanding. This cosmic microwave background (CMB) radiation was discovered (accidentally!) by Arno Penzias and Robert Wilson,¹⁵ who were awarded the 1978 Nobel prize in Physics. Over the years it has been confirmed that this radiation is almost perfectly isotropic and fits a blackbody distribution with a temperature of 2.72 K. The most accurate measurements have been obtained from the COBE satellite (Cosmic Background Explorer) launched in 1989.

Thermal radiation, resulting from a hot beginning of the Universe, had already been predicted by Gamow in 1946 and developed in key papers by Alpher, Bethe and Gamow and by Alpher and Herman in 1948.¹⁶ The model they proposed, now accepted as the ‘hot big bang’ model, is that in its early moments the Universe contained matter at a high density and

¹⁴ Hartle (2003), p. 410.

¹⁵ Penzias & Wilson (1965).

¹⁶ Alpher, Bethe & Gamow (1948), Alpher & Herman (1948a,b).

high temperature. This presupposes thermal equilibrium and therefore thermal (blackbody) radiation, with equilibrium between matter and radiation being maintained by reactions such as $p + n \leftrightarrow d + \gamma$, $e^+ + e^- \leftrightarrow \gamma + \gamma$ and so on. The idea is that as the Universe has expanded from this early phase the blackbody radiation *has retained its blackbody character* and simply cooled. To demonstrate the feasibility of this model we need to show that it is actually true – that in our model for the expansion of the Universe, blackbody radiation does indeed retain its thermal character as space expands. This is what we shall now demonstrate.

First, note that the relation between density and temperature of blackbody radiation is $\rho = aT^4$ (where $a = 4\sigma/c$ and σ is the Stefan–Boltzmann constant). Then noting from (10.146) that $\rho \propto 1/S^4$ it follows that the variation of T with S is

$$T \propto \frac{1}{S}. \quad (10.156)$$

We shall now show that this is precisely the condition for blackbody radiation to retain its thermal character under space expansion – for the Planck spectrum to be preserved. The proof is based on conservation of photon number. In the Planck distribution the number $dn(\nu)$ of photons with frequency between ν and $\nu + d\nu$ in a volume $V(t_1)$ of space at time t_1 is

$$dn(t_1) = \frac{8\pi\nu^2 V(t_1) d\nu}{c^3 \left[\exp\left(\frac{h\nu}{kT(t_1)}\right) - 1 \right]}. \quad (10.157)$$

At a later time t_2 the frequency has been red-shifted to (see (10.78))

$$\nu_2 = \nu_1 \cdot \frac{S(t_1)}{S(t_2)} \quad (10.158)$$

and the volume changed to

$$V(t_2) = V(t_1) \cdot \frac{S^3(t_2)}{S^3(t_1)},$$

hence

$$\nu_1^2 V(t_1) d\nu_1 = \nu_2^2 V(t_2) d\nu_2 \quad (10.159)$$

and the – *conserved* – number of photons in the stated frequency range is, from (10.157), using (10.158)

$$dn(t_2) = \frac{8\pi\nu_2^2 V(t_2) d\nu_2}{c^3 \left[\exp\left(\frac{h\nu_2}{kT(t_2)}\right) - 1 \right]}.$$

The condition that this is the same as $dn(t_1)$ is

$$\frac{\nu_2}{T(t_2)} = \frac{\nu_1}{T(t_1)},$$

i.e. from (10.158)

$$T(t_2) = T(t_1) \cdot \frac{S(t_1)}{S(t_2)},$$

or

$$T(t) \propto \frac{1}{S(t)}, \quad (10.160)$$

which is exactly the same as (10.156). Hence conservation of photon number guarantees that the blackbody nature of the radiation is preserved under space expansion. Finally it is useful to recall that the energy density of the radiation is, from (10.157),

$$\rho_r(t) = \int \frac{h\nu}{V(t)} dn(t) = \frac{8\pi h}{c^3} \int_0^\infty \frac{\nu^3 d\nu}{\exp\left(\frac{h\nu}{kT(t)}\right) - 1} = \frac{8\pi^5 k^4}{15c^3 h^3} T^4(t) = a T^4(t), \quad (10.161)$$

where $a = 4\sigma/c = 7.56 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$, σ is the Stefan–Boltzmann constant, and we have used the well-known integral¹⁷

$$I = \int_0^\infty \frac{z^3 dz}{e^z - 1} = \frac{\pi^4}{15}. \quad (10.162)$$

10.9 Brief sketch of the early Universe

It seems that almost everyone in the world knows about the Big Bang theory; that the Universe began in a hot and very dense phase, from which it has expanded, and is still expanding, some thousands of millions of years later. It is indeed a most remarkable achievement of twentieth-century physics; that by using the laws of nature, formulated here on Earth in the last three hundred years or so, we are able to give such a detailed account of the first few *minutes* of the Universe. In particular, we are able to ‘predict’ helium synthesis during these early moments, with very good numerical agreement with the amount of primordial helium found in the Sun and the interstellar medium. Amongst other things, this must imply that the laws of nature have *not changed with time*. This is surely a remarkable fact; the laws we understand at present are something like the absolute truth. In my opinion the Big Bang account of the birth of the Universe is so remarkable that every educated person should be aware of it, and for this reason the present section is devoted to a broad overview of its main features. For more detailed accounts the reader is referred elsewhere.

We have seen (see Fig. 10.13) that if there is any radiation present in the Universe now, in the early moments its contribution to the dynamics of its evolution must have dominated that

¹⁷ See for example Mandl (1988), Appendix A1.

of ordinary matter. And indeed there is thermal radiation in the Universe so we can proceed to investigate Equation (10.129),

$$\dot{S}^2 + kc^2 = \frac{8\pi G}{3}\rho S^2 \quad (10.129)$$

with $\rho \propto 1/S^4$, as in (10.146). It is clear that as we go backwards in time S becomes smaller and smaller, the right hand side of (10.129) becomes larger and larger, so that as $S \rightarrow 0$,

$$\frac{8\pi G}{3}\rho S^2 \gg kc^2$$

and (10.129) reduces to

$$\dot{S}^2 = \frac{8\pi G}{3}\rho S^2. \quad (10.163)$$

We see that the *geometry* of space-time is *irrelevant* in the early Universe.

We should note that ρ properly stands for the density of *all relativistic* particles, not only photons ('relativistic' means $E \gg mc^2$). For example, at a temperature of $T = 10^{10}$ K, $kT = 8.7 \times 10^5$ eV ≈ 1 MeV. But the rest mass of an electron is $mc^2 = 0.5$ MeV, so electrons are effectively relativistic at this temperature, as are also electron neutrinos ($mc^2 < 3$ eV) and muon neutrinos ($mc^2 < 0.2$ MeV), as well as the antiparticles of all these particles. We note that electrons and neutrinos are fermions, so including their contribution to ρ , the total (mass) density, means that instead of (10.161) we should write

$$\rho = N_{\text{eff}} \cdot \frac{a}{c^2} T^4, \quad (10.164)$$

where N_{eff} is the effective number of relativistic species of particle at a temperature T . It is not the actual number, since fermions and bosons contribute differently.

For electrons, which obey Fermi-Dirac statistics, the energy density per polarisation state is¹⁸

$$\rho_f = \frac{1}{8\pi^3} \int E(\mathbf{k}) f(E(\mathbf{k})) d\mathbf{k},$$

where $E(\mathbf{k})$ is the energy of an electron with wave vector $\mathbf{k}$ and $f(E(\mathbf{k}))$ is the Fermi function

$$f(E(\mathbf{k})) = \frac{1}{e^{E/kT} + 1} = [\exp(E/kT) + 1]^{-1}$$

(where the chemical potential has been put equal to 0). Performing the angular integrations

$$\rho_f = \frac{1}{2\pi^2} \int E k^2 [\exp(E/kT) + 1]^{-1} dk.$$

Now $k = (2\pi/\hbar)p$ and in the *relativistic* regime $E = pc$, so

¹⁸ See for example Ashcroft & Mermin (1976), p. 43. Note that the factor $1/4\pi^3$ in Ashcroft & Mermin refers to *two* polarisation states.

$$\rho_f = \frac{4\pi c}{h^3} \int_0^\infty \frac{p^3 dp}{\exp(pc/kT) + 1} = \frac{4\pi k^4}{h^3 c^3} I_1 \cdot T^4 \quad (10.165)$$

with

$$I_1 = \int_0^\infty \frac{z^3 dz}{e^z + 1}. \quad (10.166)$$

By noting that

$$\frac{2}{e^{2z} - 1} = \frac{1}{e^z - 1} - \frac{1}{e^z + 1}$$

it is easy to show that

$$I_1 = \frac{7}{8} I = \frac{7}{120} \pi^4$$

from (10.162), so the energy density per polarisation state of a relativistic fermion is

$$\rho_f = \frac{7}{16} \sigma T^4, \quad (10.167)$$

to be compared with (10.161) for photons.

To return to the early Universe, at a temperature of about 10^{10} K the electron and the positron and the electron and muon neutrinos and their antiparticles will all be relativistic:

$$e^-, e^+, \nu_e, \bar{\nu}_e, \nu_\mu, \bar{\nu}_\mu.$$

In the usual theory of beta decay, the observed parity violation is ascribed to the fact that ν_e is massless and exists in only one polarisation state (a left-handed state). By an extension of ideas the same is supposed to be true of ν_μ ; whereas by contrast e^- and e^+ have $2 (= 2s + 1)$ polarisation states so the effective number of relativistic particles is

$$N_{\text{eff}} = 1 + [2 \times 2 + 4 \times 1] \cdot \frac{7}{16} = \frac{9}{2} = 4.5. \quad (10.168)$$

In recent years, however, the theory of *neutrino oscillations* has received much attention since it offers a solution to the solar neutrino problem. The general idea of neutrino oscillations is that while ν_e are emitted from the Sun, for example as products of boron decay ${}^8\text{B} \rightarrow {}^8\text{Be} + e^+ + \nu_e$, by the time these neutrinos reach the Earth they have become a mixture of ν_e , ν_μ and ν_τ ; and only the electron neutrinos will be detectable in the laboratory, resulting in a reduced detection rate; this smaller than expected detection rate is the solar neutrino problem, which is therefore solved in this theory of oscillations. The theory, however, only works if the electron and muon neutrinos have different masses (the effect depends on the difference in their $(\text{mass})^2$), so they cannot both be massless. Assuming, for the sake of definiteness, that neither of them is massless, they will possess two polarisation states each and instead of (10.167) we will have

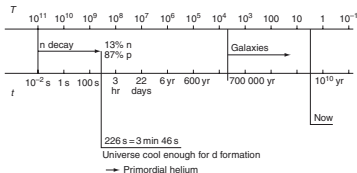


Fig. 10.16

Thermal history of the Universe.

$$N_{\text{eff}} = 1 + [2 \times 2 + 4 \times 2] \cdot \frac{7}{16} = \frac{25}{4} = 6.25. \quad (10.169)$$

In any case, substituting (10.163) into (10.164) gives

$$\frac{1}{S^2} \left(\frac{dS}{dt} \right)^2 = \frac{8\pi G a}{3c^2} \cdot N_{\text{eff}} \cdot T^4 \equiv b^2 T^4. \quad (10.170)$$

We want an equation relating T and t . Equation (10.160), however, implies that

$$\frac{1}{S^2} \left(\frac{dS}{dt} \right)^2 = \frac{1}{T^2} \left(\frac{dT}{dt} \right)^2,$$

so (10.170) gives

$$\left(\frac{dT}{dt} \right)^2 = b^2 T^6 \Rightarrow \frac{dT}{dt} = -b T^3, \quad (10.171)$$

where the minus sign is chosen since T decreases as t increases. Integrating (10.171) gives

$$T\sqrt{t} = \frac{1}{\sqrt{2b}} = 1.52 N_{\text{eff}}^{-1/4} \times 10^{10} \text{ s}^{1/2} \text{ K}.$$

With $N_{\text{eff}} = 4.5$ this gives

$$T\sqrt{t} \sim 1.04 \times 10^{10} \text{ s}^{1/2} \text{ K},$$

or

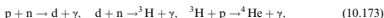
$$T\sqrt{t} \approx 10^{10}. \quad (10.172)$$

This relates the temperature of the radiation in K to the age of the Universe in seconds, and it crucially allows us to construct a *thermal history of the Universe*, at least in its early stages.

Equation (10.172) is displayed in Fig. 10.16, where both T and t are plotted horizontally on a logarithmic scale. When the universe is 1 second old the temperature is 10^{10} degrees,

when $t = 100$ s the temperature has dropped to 10^9 K and so on. It is interesting to see that if one inserts the present age, $t \sim 10^{18}$ s, into (10.172) one finds $T = 10$ K, not too far from the actual value of 2.7 K. But (10.172) was derived for a *radiation-dominated universe*, which the present Universe is *not*, so there is no reason to expect (10.172) to work all that well for the Universe at the present time.

What was the Universe like at early times? If we take a journey backwards in time we see the Universe contract and heat up. When T reaches about 10^4 K, kT is about 1 eV, of the order of atomic ionisation energies, so at this temperature matter, composed of atoms and molecules, will lose its structure and be reduced to atomic nuclei and electrons. When T reaches 10^{10} degrees the thermal energy of the order of MeV is enough to dissociate nuclei, so above this temperature the Universe will consist of protons, neutron, photons, muons, electrons and neutrinos (and antiparticles of these last three). All structure will have disappeared. The Universe is a simple mixture of these particles at a very high temperature. Coming forwards in time, when T drops to about 1000 K the plasma will condense into electrically neutral atoms, so electromagnetic forces will disappear on the large scale and gravity will take over, allowing for galactic condensation, and, in due course, condensation into stars, in which nuclear fusion reactions take place, to produce the chemical elements. The first few of these reactions are



and these result in helium production. Heavier nuclei are produced in analogous reactions. Given our picture of the early Universe, however, since it consists of protons and neutrons an obvious question is: did the reactions above take place then, producing helium? And if so, how much helium? And were heavier elements produced? Let us consider the first of these questions.¹⁹

When $t = 0.01$ s, $T = 10^{11}$ K and at that temperature the number of protons is the same as the number of neutrons, $N(p) = N(n)$. The argument goes as follows. The following reactions convert n into p and vice versa:



so under conditions of thermal equilibrium the relative population of proton and neutron states will obey a Maxwell–Boltzmann distribution

$$\frac{N(n)}{N(p)} = \exp(-\Delta E/kT), \quad (10.174)$$

where $\Delta E = [m(n) - m(p)]c^2 = 1.3$ MeV. When $T = 10^{11}$ K, however, $kT = 8.6$ MeV and the above ratio is very close to 1, so at an age of 1/100 second the proton/neutron balance is 50% n , 50% p . Thereafter however, as T drops, an equal population of proton and neutron levels is not guaranteed and the neutrons will start to decay



¹⁹ I must emphasise once more that this account is greatly simplified, since I am aiming only to give the broadest of outlines of this topic. Much more detail will be found in the texts cited in ‘Further reading’.

The question we are asking is whether the fusion reactions (10.173) will take place in the environment of the early Universe. The answer is that they will not produce stable helium in a hot universe, for the photons will still have enough energy to destroy the deuterium as fast as it is created

$$p + n \leftrightarrow d + \gamma.$$

We have to wait until the photons are not energetic enough for this reaction to go ‘backwards’. This energy is easily measured in a laboratory and corresponds to a temperature of about 0.7×10^9 K, which in turn corresponds to a time of 226 seconds. After this time, the ‘deuterium bottleneck’ has been passed and the further reactions in (10.173) will proceed, producing ${}^4\text{He}$. During these 226 seconds, however, the neutrons have been decaying and the 50% p, 50% n mixture at $t = 0.01$ seconds has now become, at 226 seconds, 87% p, 13% n. The fusion reactions which then go ahead produce $2 \times 13 = 26\%$ ${}^4\text{He}$ and 74% H; and it is *this* mixture of hydrogen and helium (with a small admixture of heavier elements) which, hundreds of thousands of years later, will constitute the gas forming the raw material of stars and the interstellar medium.

This, in the broadest of terms, is how the ‘standard model’ of Big Bang Cosmology predicts the primordial synthesis of helium (as well as some heavier elements). The agreement between the theoretical prediction and the observed data is impressive.²⁰ For example, in HII regions of the interstellar medium²¹ the proportion of ${}^4\text{He}$ to all nucleonic matter is predicted to be $Y \approx 0.23$ and is found²² to be

$$Y = 0.231 \pm 0.006.$$

It should be remarked that one surprising feature of this model is its *simplicity*. It depends on a homogeneous universe, Einstein’s field equations applied to a radiation gas, and the well-established fields of thermodynamics and nuclear physics. No new laws of physics have had to be invented. This is surely a remarkable fact – that we can account for the evolution of the Universe from the time when it was only 1/100 second old! From a rather different perspective, what is also remarkable is that the original Big Bang model of Alpher and Herman was for a long time not taken seriously. In fact it was not until the 1960s, when quasars and pulsars were discovered and when in particular John Wheeler (1911–2008) began to urge the theoretical community to take seriously the work of Oppenheimer and his collaborators on gravitational collapse and, in the words of Freeman Dyson, to ‘rejuvenate General Relativity; he made it an experimental subject and took it away from the mathematicians’,²³ that physicists seriously began to take on board the idea that there actually was a time when the Universe was only a few minutes old. In the words of Weinberg:²⁴

This is often the way it is in physics – our mistake is not that we take our theories too seriously, but that we do not take them seriously enough. It is always hard to realise that these numbers and equations we play with at our desks have something to do with the real

²⁰ For a recent review, see for example Hogan (1997).

²¹ Regions of ionised hydrogen: see for example Carroll & Ostlie (1996).

²² Skillman & Kennicutt (1993).

²³ *New York Times*, 14 April 2008.

²⁴ Weinberg (1978), p. 128.

world. Even worse, there often seems to be a general agreement that certain phenomena are just not fit subjects for respectable theoretical and experimental effort. Gamow, Alpher and Herman deserve tremendous credit above all to take the early universe seriously, for working out what known physical laws have to say about the first three minutes. Yet even they did not take the final step, to convince the radio astronomers that they ought to look for a microwave radiation background. The most important thing accomplished by the ultimate discovery of the 3 °K radiation background in 1965 was to force us all to take seriously the idea that there *was* an early universe.

10.10 The inflationary universe and the Higgs mechanism

Despite the outstanding success of the Big Bang model there are still some questions left unanswered. Prominent among these is the high degree of *isotropy* of the cosmic background radiation. The temperature (2.72 K) is the same in all directions – including diametrically opposite ones. This creates a difficulty because of the horizon problem, described above. The radiation reaching our apparatus has travelled freely through the Universe since the recombination era, at about $t = 130\,000$ years. (This is a recombination of protons and electrons into hydrogen, and happens when the radiation is at a temperature T such that kT is rather less than the ionisation energy of hydrogen (13.6 eV), since even at that energy the blackbody spectrum means that in the Wien tail (the high energy end of the spectrum) there will still be enough photons to ionise hydrogen.) The radiation was *thermalised*, however, much earlier, during the first few seconds. Referring to Fig. 10.7, consider a journey backwards in time. As we approach the Big Bang we see that the horizons of a given world-line shrink – other bodies in space become progressively invisible as the light from them has not yet had time to reach us. At the present time we are able to see the light from very many galaxies, but at early times these galaxies – particularly ones at ‘opposite ends’ of the Universe – must have been definitely *not* within each other’s light cone. On the other hand the cosmic background radiation from these directions is at the *same* temperature, which surely argues that these regions must have been in causal contact at very early times, and therefore *within* each other’s light cones.

This is known as the *horizon problem*. A solution was proposed by Guth in 1981:²⁵ all the points in space – even those in diametrically opposite directions – from which we are now receiving light *were* actually *within* each other’s light cones at very early times when thermal equilibrium held, and since that time space has expanded by such an enormous factor that these regions now make up the whole of the visible Universe. The situation is sketched in Fig. 10.17. Let us calculate what expansion factor is necessary for this purpose. Guth’s argument was developed in the context of particle physics and the reasoning is explained in some detail below. For reasons to be explained there, the expansion is reckoned to have begun at a time related to the ‘Grand Unification’ energy

$$E_{\text{GUT}} \approx 10^{14} \text{ GeV}, \quad (10.176)$$

²⁵ Guth (1981).

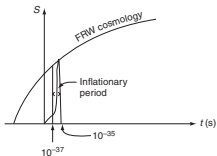


Fig. 10.17

The inflationary phase of expansion, starting at about 10^{-37} seconds and finishing at about 10^{-35} seconds.

which corresponds to a temperature of $T_{\text{GUT}} \approx 10^{27}$ K and therefore to a time, from (10.172)

$$t_{\text{GUT}} \approx 10^{-34} \text{ s.} \quad (10.177)$$

At this very early time the size of a causally connected region of space is $\sim ct_{\text{GUT}} \approx 10^{-26}$ m. If the expansion of the Universe since that early time had simply followed a Friedmann–Robertson–Walker pattern for a radiation-dominated universe, the ratio of the present value of the expansion factor S_0 to its value at the GUT time, S_{GUT} , is, from (10.156) and (10.172) and noting that the present age of the Universe $\approx 10^{17}$ s,

$$\frac{S_0}{S_{\text{GUT}}} \approx \frac{T_{\text{GUT}}}{T_0} \approx \left(\frac{t_0}{t_{\text{GUT}}} \right)^{1/2} \approx \left(\frac{10^{17}}{10^{-34}} \right)^{1/2} \approx 10^{26}. \quad (10.178)$$

Under this expansion the causally connected region, originally of size $\sim 10^{-26}$ m would now be of size ~ 1 m. Guth's hypothesis, however, is that it should have expanded to the size of the visible Universe, that is to $\sim c \times 10^{17} \approx 10^{26}$ m, which represents a *further* expansion by a factor $\sim 10^{26}$:

$$\text{inflationary factor} \approx 10^{26}. \quad (10.179)$$

Guth's model, known as the *inflationary universe*, is that for a very short time, beginning at t_{GUT} above, the Universe suffered this incredibly large expansion. Exactly how long the expansion lasted (perhaps $100 \times t_{\text{GUT}}$) and, crucially, how the inflationary period *ended* (the so-called 'graceful exit' problem) are more detailed matters which we shall not consider here. But the general idea is that after a (very) short time the expansion of space settled down to the extremely sedate expansion of the Friedmann–Robertson–Walker cosmology that we see now.

The physical motivation for the inflationary model comes from particle physics, in particular the notions of *spontaneous symmetry breakdown* and the *Higgs mechanism*. The key development was the theory of unification of the electromagnetic and weak interactions, into the so-called electroweak interaction. The prototype weak interaction is neutron beta decay

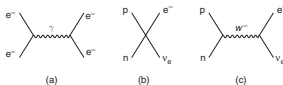


Fig. 10.18

Feynman diagrams for (a) electron–electron interaction via photon exchange, (b) $n + \nu_e \rightarrow p + e^-$ in Fermi's theory, with no field, (c) the same reaction mediated via W exchange.

$$n \rightarrow p + e^- + \bar{\nu}_e.$$

The (quantum field theoretic) amplitude for this decay is essentially the same as that for the related scattering process

$$\nu_e + n \rightarrow e^- + p. \quad (10.180)$$

We may set this alongside the electromagnetic process

$$e^- + e^- \rightarrow e^- + e^- \quad (10.181)$$

of electron–electron scattering. Electrons interact with each other through the electromagnetic field, and, at the quantum level, the basic interaction is represented by the Feynman diagram in Fig. 10.18(a) – two electrons exchange one photon, a quantum of the electromagnetic field.

According to Fermi's theory of weak interactions, on the other hand, the Feynman diagram for the process (10.180) is drawn in Fig. 10.18(b). This is a *contact interaction*. The wave functions for the four particles ν_e , n , e^- and p must all overlap at one point. There is no action-at-a-distance, therefore no need for a field, therefore no field responsible for this weak interaction. Fermi's theory, however, suffered from convergence problems at higher orders in perturbation theory (basically it is 'non-renormalisable'), but these problems are made much less severe if it is supposed that the weak interaction, like the electromagnetic one, is mediated by a *field*. In that case the Feynman diagram of Fig. 10.18(b) is replaced by Fig. 10.18(c); here W^- is the quantum of the weak field. This is the basic motivation behind electroweak theory, and in that theory there are actually four field quanta: γ , W^- , W^+ (its antiparticle) and Z^0 . The photon is massless but the others have masses,

$$m_W = 81.8 \text{ GeV}/c^2 = 87 m_p, \quad m_Z = 92.6 \text{ GeV}/c^2 = 99 m_p \quad (10.182)$$

(m_p = proton mass). The W and Z particles were predicted at these masses, and found at CERN in 1983.

It is the fact that the masses of W and Z are non-zero that creates the problem, for electroweak theory is in essence a generalisation of Maxwell's electrodynamics, from a gauge group $U(1)$ to the non-abelian group $SU(2) \otimes U(1)$; and in that generalisation the new gauge fields would, like the photon, all have zero mass. The mechanism which allows the W and Z fields to be massive (but γ to remain massless) is known as the *Higgs mechanism*.

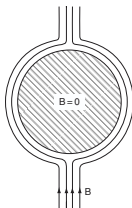


Fig. 10.19

The Meissner effect. Magnetic flux is expelled from a superconductor.

Higgs' work²⁶ concerned the masses of gauge particles in theories with so-called spontaneously broken symmetry, and a good illustration of this, one in which Higgs was himself interested, is the phenomenon of superconductivity.

The defining characteristic of superconductivity is that at a temperature below a *critical temperature* T_c , some metals lose all electrical resistance, $R=0$: the resistance is not simply very small, it is *zero*! R is defined as being the proportionality between the electric field $\mathbf{E}$ and electric current $\mathbf{j}$:

$$\mathbf{E} = R \mathbf{j}, \quad (10.183)$$

or equivalently

$$\mathbf{j} = \sigma \mathbf{E} \quad (10.184)$$

(where σ is the conductivity). A metal in a superconducting state then exhibits a persistent current even in no field: $\mathbf{j} \neq 0$ when $\mathbf{E} = 0$. The key to understanding superconductivity is to describe the current as a 'supercurrent' $\mathbf{j}_s$, and, in contrast to the equation above, to suppose that this is proportional, not to $\mathbf{E}$ but to the vector potential $\mathbf{A}$:

$$\mathbf{j}_s = -k^2 \mathbf{A}, \quad (10.185)$$

with a negative coefficient of proportionality. This is the *London equation*. In a static situation, even if $\mathbf{A} \neq 0$, $\mathbf{E} = -\partial \mathbf{A} / \partial t = 0$, so if R is defined by $\mathbf{E} = R \mathbf{j}_s$, then $R = 0$, as required.

For our purposes the crucial property of superconductors is the *Meissner effect*, which is the phenomenon that magnetic flux is expelled from superconductors, as sketched in Fig. 10.19 – an external magnetic field does not penetrate a superconductor. Higgs' contribution was to show that, suitably transformed into a relativistic theory, this is equivalent to

²⁶ Higgs (1964a, b, 1966).

saying that the photon has an effective mass. The reasoning goes as follows. First, the London equation explains the Meissner effect; for taking the curl of Ampère's equation

$$\nabla \times \mathbf{B} = \mathbf{j}$$

gives

$$\nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} = \nabla \times \mathbf{j}.$$

Applying this to the supercurrent (10.185) and noting that $\nabla \cdot \mathbf{B} = 0$ (no magnetic monopoles) gives

$$\nabla^2 \mathbf{B} = k^2 \mathbf{B}. \quad (10.186)$$

In one dimension the solution to this is

$$B(x) = B(0) \exp(-kx), \quad (10.187)$$

which describes the Meissner effect – the magnetic field is exponentially damped inside the superconductor, only penetrating to a depth of order $1/k$.

Equation (10.186) is equivalent to

$$\nabla^2 \mathbf{A} = k^2 \mathbf{A}. \quad (10.188)$$

This equation is, however, non-relativistic. To make it consistent with (special!) relativity ∇^2 should be replaced by the Klein–Gordon operator $\square$ and $\mathbf{A}$ by the 4-vector $A^\mu = (\phi, \mathbf{A})$ (as in (2.85) above), giving

$$\left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) A^\mu = k^2 A^\mu. \quad (10.189)$$

The vector potential is a ‘field’ quantity, but we are interested in the photon, the quantum of the field, so we make the transition to quantum theory by the usual prescription

$$\frac{\partial}{\partial t} \rightarrow -\frac{i}{\hbar} E, \quad \frac{\partial}{\partial x} \rightarrow \frac{i}{\hbar} p_x, \text{ etc.},$$

giving, for the quantum of the field A^μ

$$E^2 - p^2 c^2 = k^2 c^2 \hbar^2, \quad (10.190)$$

where E is the (total, including rest-mass) energy of the field quantum and p its momentum. Comparison with Einstein's relation $E^2 - p^2 c^2 = m^2 c^4$ implies that the mass of the quantum in a superconductor is

$$m_\gamma = \frac{k\hbar}{c}; \quad (10.191)$$

the photon behaves as (in effect *is*) a massive particle. This is the import of the Meissner effect.

All this would seem to be a long way from cosmology and the early Universe, but the connection is made by appealing to the Bardeen–Cooper–Schrieffer (BCS) theory of

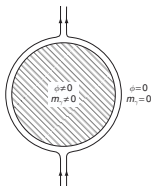


Fig. 10.20

Photons are massive in a superconductor, where the scalar (BCS) field ϕ is non-zero, but massless outside, where there is no field.

superconductivity,²⁷ which is a microscopic theory. At the quantum level the BCS theory accounts for superconductivity by positing a *scalar field* ϕ (i.e. of spin 0) which describes a ‘Cooper pair’ of electrons (e^-e^-); the pairing is in momentum space rather than coordinate space. Combining this idea with the previous reasoning, we now have the situation that superconductivity is described by a *many-particle wave function (or field)* ϕ , non-zero inside a superconductor, in which also $m_\gamma \neq 0$. Outside the superconductor $\phi = 0$ and $m_\gamma = 0$. This is shown in Fig. 10.20. When a superconductor is *heated* to above the critical temperature, of course it loses its superconductivity, reverting to a ‘normal’ state. In this transition the field $\phi \rightarrow 0$ and the photon becomes massless. This is a thermodynamic phase transition.

A few years after the work of Higgs, and in the context of elementary particle physics, Weinberg²⁸ and Salam²⁹ proposed a unified model of weak and electromagnetic interactions. In this model it is necessary that the weak field quanta be massive (the reason is essentially that the weak force is of extremely short range, and is therefore carried by massive particles³⁰). On the other hand, as has already been remarked, the unified electroweak theory is ‘simply’ a generalisation of Maxwell’s electrodynamics, so the field quanta should all have zero mass. This problem is solved by invoking a mechanism *analogous* to superconductivity. There is a scalar field – now called the Higgs field – which is all-pervasive (unlike the BCS field, which only exists in a superconductor). By virtue of this field the W and Z particles acquire a mass, in the way that the photon acquires a mass in superconductivity, but it can be arranged (somewhat miraculously!) that the photon remains massless. In the Weinberg–Salam theory the Higgs field is a complex scalar field but to illustrate the mechanism at work we may consider a real field ϕ with a potential energy function (sometimes called the Higgs potential)

²⁷ Bardeen, Cooper & Schrieffer (1957).

²⁸ Weinberg (1967).

²⁹ Salam (1968).

³⁰ The range is proportional to the inverse of the mass of the field quantum; for example electrodynamics has ‘infinite’ range and the photon is massless.

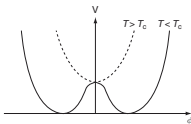


Fig. 10.21

The Higgs potential. When $T < T_c$ it has a maximum at $\phi = 0$ and two minima at $\phi = \pm a$. When $T > T_c$ there is only a minimum at $\phi = 0$.

$$V(\phi) = \frac{m^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4. \quad (10.192)$$

On quantisation m becomes the particle mass and the ϕ^4 term represents a quartic self-interaction. The extremal values of $V(\phi)$, given by $\partial V/\partial \phi = 0$ (differentiation with respect to the function ϕ), are clearly

$$\phi = 0, \pm \sqrt{\frac{-m^2}{\lambda}} \equiv 0, \pm a. \quad (10.193)$$

In a usual field theory $m^2 > 0$ and the second two possibilities are non-physical but if we are prepared to consider the case $m^2 < 0$ then the function $V(\phi)$ has two minima and a maximum, as shown in Fig. 10.21. The maximum is at $\phi = 0$ and the minima at $\phi = \pm a$. A potential function like this begs a question about the nature of the vacuum. Normally the term ‘vacuum’ has two connotations: it is both the state of lowest energy of a system, and also the state in which no matter and no fields are present. Here, however, these situations become *distinct*: when there is no field, $\phi = 0$, the energy is *not* a minimum, but a maximum; and the state of lowest energy is a state in which the field does not vanish (and is also 2-fold degenerate).

In the case of superconductivity ϕ would represent the BCS condensate, so that $\phi \neq 0$ (and $m_\gamma \neq 0$) when $T < T_c$; but on heating the material to a temperature $T > T_c$, $\phi = 0$ and m_γ also becomes 0. The same assumption is made in particle physics: in the actual world the Higgs field ϕ does not vanish (its vacuum value is non-zero) and the quanta W and Z are massive. But if the temperature were to rise to a value that we may call $T_{\text{el.wk}}$ (the subscript standing for ‘electroweak’) given by (see (10.176))

$$kT_{\text{el.wk}} = m_Z c^2 \Rightarrow T_{\text{el.wk}} \approx 10^{15} \text{ K}, \quad (10.194)$$

then the vacuum would shift to $\phi = 0$. At temperatures higher than this W, Z and γ would all be massless and the electroweak symmetry becomes an *exact* symmetry; at lower temperatures it is referred to as a *spontaneously broken symmetry*. This, then, would be the situation in the early Universe, at times earlier than (see (10.172))

$$t_{\text{el.wk}} = \left(\frac{10^{10}}{T_{\text{el.wk}}} \right)^2 \approx 10^{-10} \text{ s}. \quad (10.195)$$

At these times, when $T > T_{\text{el.wk}}$, the vacuum is a 'true' minimum of $V(\phi)$, at $\phi = 0$. As the temperature decreases to below $T_{\text{el.wk}}$ this becomes a 'false vacuum', and ϕ then 'rolls down' to the new true vacuum at $\phi = \pm a$. As a result of this phase transition the particles W and Z pick up a mass.

The final connection with the cosmology of the early Universe is made by making an identification between the field ϕ and the cosmological constant Λ , as first suggested by Zeldovich.³¹ The consequence of this is that when the vacuum expectation value of ϕ is non-zero, the cosmological constant dictates the dynamics, the Universe behaves like a de Sitter universe and *expands exponentially*. So the situation runs along the following lines: at very early times, when the temperature is extremely high (in the present reasoning, higher than 10^{15} K, as above), electroweak symmetry is exact, all field quanta are massless and the vacuum value of the Higgs field is zero – as shown in Fig. 10.21. The expansion parameter goes as $t^{2/3}$ (see Problem 10.3). When the temperature drops to below $T_{\text{el.wk}}$ the symmetry is broken, the W and Z quanta become massive and the vacuum value of the scalar field ϕ is non-zero. The Higgs field, in the shape of a cosmological constant, then drives the expansion, which is exponential. This lasts for a short time, after which the expansion reverts to a 'sensible' Friedmann–Robertson–Walker type, and the horizon problem is solved. It should be noted that the *flatness problem* is also solved. It was noted above that an analysis of the inhomogeneities of the CMB radiation spectrum indicated that the Universe is flat, or almost flat. A priori this is difficult to account for, but if the Universe has undergone an inflationary phase then this will result in space becoming almost flat; the more a balloon is inflated the smaller the curvature of its surface becomes.

Finally, to give an account of Guth's model, some adjustments have to be made to the picture above. The first one is that the temperature $T_{\text{el.wk}}$ which we have been dealing with is not the temperature assumed in the literature. There the assumption is made that there exists a 'Grand Unified Theory' (GUT) of particle interactions. This goes one better than the unification of electromagnetic and weak interactions achieved by Weinberg and Salam, to embrace QCD, the theory of the strong interaction between quarks, whose quanta are gluons, so this is a unification of strong, electromagnetic and weak interactions. In this theory there are additional field quanta, much heavier than W and Z, so the temperature at which they would become massless is correspondingly higher than $T_{\text{el.wk}}$, and the time correspondingly earlier than $t_{\text{el.wk}}$. The typical GUT energy is of the order of 10^{14} GeV, corresponding to a temperature of 10^{27} K and a time of about 10^{-34} s, so this would mark the onset of inflation. In honesty it should be pointed out, however, that the status of GUTs is not particularly assured. There are actually several variants of GUT, but the simplest one predicts proton decay

$$p \rightarrow e^+ + \pi^0$$

with an estimated half-life of $10^{30 \pm 1}$ years (though this is model dependent). The experimental figure is $> 5 \times 10^{32}$ years.³² It may be that nature does not recognise a unification of interactions at this level.

³¹ Zeldovich (1968).

³² The reader might wonder at these figures, which far outstrip the age of the Universe. But recall that the decay of particles obeys a probability law, so a lifetime of 10^{30} years means that in a sample of 10^{30} protons (a decent mountain) one will decay each year.

Further reading

A superb account of cosmology, at a largely non-mathematical level, is Harrison (2000). Treatments of Killing vectors and isometries in their relation to cosmology will be found in Robertson & Noonan (1968), Weinberg (1972), Ryan & Shepley (1975), Ciufolini & Wheeler (1995), McGlinn (2003) and Plebański & Krasinski (2006). A somewhat older, though still interesting, account may be found in Landau & Lifshitz (1971). A very nice introduction to Gaussian curvature and the differential geometry of curves and surfaces is Faber (1983); see also Struik (1961) and Stoker (1969).

A very good account of horizons is Rindler (2001), Section 17.3. Detailed accounts of the luminosity–red-shift relation may be found in McVittie (1965), Weinberg (1972) and Peebles (1993). For further information about dark matter see Coles & Ellis (1997), Peacock (1999), Harrison (2000), Hartle (2003) and Roos (2003).

Detailed treatments of the early Universe may be found in Weinberg (1972), Börner (1988), Kolb & Turner (1990), Peebles (1993), Coles & Ellis (1997) and Mukhanov (2005). A full account of neutrino oscillations may be found in Raffelt (1996).

Good introductory accounts of superconductivity can be found in Feynman *et al.* (1965), Chapter 21 and Ashcroft & Mermin (1976), Chapter 34. For the Higgs mechanism and its role in electroweak theory see Aitchison & Hey (1982), Cao (1997), Cottingham & Greenwood (1998), Rubakov (2002), Maggiore (2005), Srednicki (2007), Chapters 9 and 10, or Ryder (1996). A particularly attractive and authoritative (and almost non-mathematical) account is Taylor (2001). Detailed accounts of inflationary cosmology appear in Börner (1988), Kolb & Turner (1990), Linde (1990), Mukhanov (2005) and Hobson *et al.* (2006). Recent reviews of the inflationary universe may be found in Guth (2000) and Linde (2000). A good critique of the inflationary model is Penrose (2004), Chapter 28.

Problems

- 10.1 Prove that the curvature scalar of S^3 is $R = 6/a^2$.
- 10.2 Prove that in the Steady State Theory there is an event horizon but no particle horizon.
- 10.3 Show that for a radiation-dominated universe $S \propto t^{1/2}$ and for a matter-dominated one $S \propto t^{2/3}$.